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Blastocyst-like structures generated from human pluripotent stem cells

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Limited access to embryos has hampered the study of human embryogenesis and disorders that occur during early pregnancy. Human pluripotent stem cells provide an alternative means to study human development in a dish[1–7]. Recent advances in partial embryo models derived from human pluripotent stem cells have enabled human development to be examined at early post-implantation stages[8–14]. However, models of the pre-implantation human blastocyst are lacking. Starting from naive human pluripotent stem cells, here we developed an effective three-dimensional culture strategy with successive lineage differentiation and self-organization to generate blastocyst-like structures in vitro. These structures—which we term 'human blastoids'—resemble human blastocysts in terms of their morphology, size, cell number, and composition and allocation of different cell lineages. Single-cell RNA-sequencing analyses also reveal the transcriptomic similarity of blastoids to blastocysts. Human blastoids are amenable to embryonic and extra-embryonic stem cell derivation and can further develop into peri-implantation embryo-like structures in vitro. Using chemical perturbations, we show that specific isozymes of protein kinase C have a critical function in the formation of the blastoid cavity. Human blastoids provide a readily accessible, scalable, versatile and perturbable alternative to blastocysts for studying early human development, understanding early pregnancy loss and gaining insights into early developmental defects.

Human pluripotent stem cells (PSCs) that are derived from pre-implantation embryos (embryonic stem cells, ES cells) or obtained through cellular reprogramming (induced pluripotent stem cells, iPS cells) provide an invaluable in vitro model to study early human development, and hold great promise for regenerative medicine[1–7]. Recent progress in stem-cell-based embryo models has opened new possibilities for research using human PSCs. Although various structures that mimic specific aspects of early human post-implantation development have been generated[8–14], models that recapitulate embryogenic events of the human blastocyst are not available. Stem-cell-derived blastocyst-like structures have been generated in mice, and these can be used to model several pre- and peri-implantation developmental processes in vitro[15–18]. However, whether human blastocyst-like structures can be generated from cultured stem cells remains unknown.

Plasticity of naive human ES cells

Recent studies have shown that naive human ES cells are competent for differentiation into both embryonic (epiblast) and extra-embryonic (trophectoderm and hypoblast) lineages[19–21]. To confirm this bipotency of naive human ES cells, we first exposed WIBR3 human ES cells, cultured in 5i/L/A medium[22], to human trophoblast stem cell medium (TSM)[19,23] and observed a gradual increase in the number of GATA3$^+$ cells (trophectoderm or trophoblast-like cells; TLCs), whereas few GATA6$^+$ cells (hypoblast-like cells, HLCs) were present (Extended Data Fig. 1a). It has been shown that activation of transforming growth factor-β (TGFβ)–SMAD and WNT–β-catenin signalling drives the differentiation of naive human ES cells towards the hypoblast lineage in vitro, and this differentiation is also dependent on fibroblast growth factor (FGF) signalling[20]. Thus, we treated WIBR3 cells with a combination of FGF2, activin A and a WNT activator (CHIR99021) (denoted as hypoblast differentiation medium, HDM), which resulted in a progressive increase in the number of GATA6$^+$ cells, but few GATA3$^+$ cells (Extended Data Fig. 1a). In addition, SOX2$^+$ cells (epiblast-like cells, ELCs) were still present after seven days of differentiation in both TSM and HDM (Extended Data Fig. 1a). Next, to enable the codifferentiation of ELCs, TLCs and HLCs, we sequentially treated WIBR3 cells with HDM followed by TSM, or TSM followed by HDM. Both methods resulted in the appearance of GATA3$^+$, GATA6$^+$ and SOX2$^+$ cells in tandem (Extended Data Fig. 1b, c). These results are consistent with previous studies that

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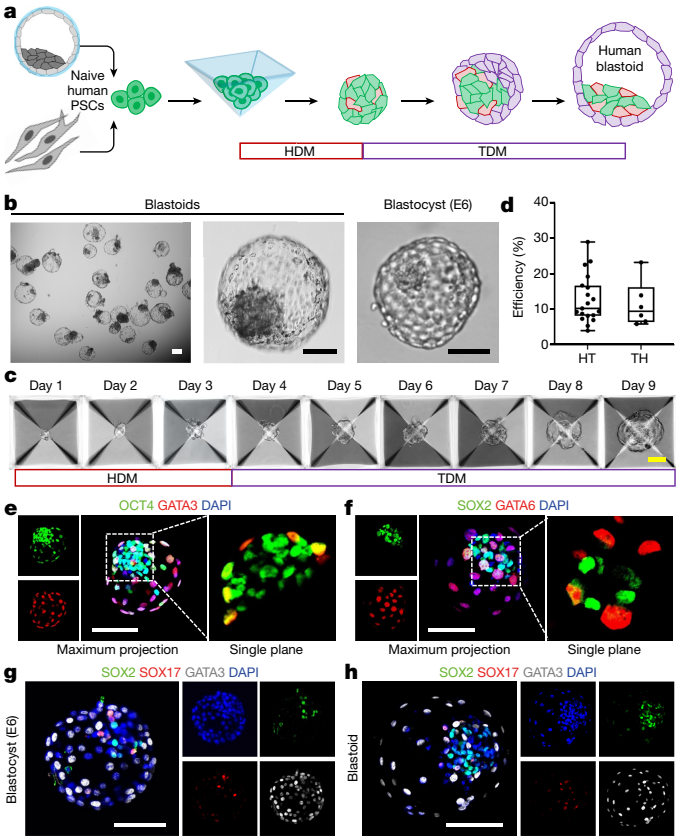

Fig. 1 | Generation of human blastocyst-like structures from naive human PSCs. a, Schematic of human blastoid formation from naive human PSCs (HT method). **b**, Representative phase-contrast images of human blastoids (left, HT method) and an E6 human blastocyst (right). Scale bars, 100 µm. **c**, Representative phase-contrast images of cell aggregates at indicated time points during blastoid formation (HT method). Scale bar, 100 µm. **d**, Blastoid formation efficiency from 5i/L/A-cultured WIBR3 human ES cells (HT method, $n = 19$; TH method, $n = 6$ (biological replicates)). Box plots show median (centre line), 25th and 75th percentiles (bottom and top of box, respectively), and minimum and maximum values (bottom and top whisker, respectively). **e**, **f**, Representative immunofluorescence co-staining images of GATA3 and OCT4 (**e**) and GATA6 and SOX2 (**f**) in human blastoids. Higher-magnification images of selected single planes of the boxed areas are shown on the right. Scale bars, 100 µm. **g**, **h**, Representative immunofluorescence co-staining images of SOX2, SOX17 and GATA3 in an E6 human blastocyst (**g**) and a human blastoid (**h**). Scale bars, 100 µm. Images in **b**, **c**, **e**–**h** are representative of at least three independent experiments.

showed plasticity of naive human ES cells towards both embryonic and extra-embryonic cell types in vitro[19–21].

Generation of human blastoids

The ability of naive WIBR3 human ES cells to differentiate into cells resembling all cell types in a blastocyst led us to test whether they can generate structures mimicking human blastocysts in vitro. To this end, we switched to three-dimensional (3D) culture using AggreWell plates coupled with sequential HDM and TSM treatments, but failed to generate cavity-containing structures. Next, we mixed TSM with 5i/L/A medium at a 1:1 ratio (denoted as trophoblast differentiation medium, TDM), and found that sequential treatment of WIBR3 cells with either HDM followed by TDM (HT method) or TDM followed by HDM (TH method) generated blastocyst-like structures containing inner cell mass (ICM)-like and trophectoderm-like compartments with a visible cavity; we term these structures 'human blastoids' (Fig. 1a–c, Extended Data Fig. 1d–f). On the basis of these morphological criteria, we quantified the blastoid formation efficiency (Fig. 1d). In addition to 5i/L/A-cultured WIBR3 cells (human ES cells), we were able to generate blastoids from 5i/L/A-cultured human iPS cells[24,25] and from WIBR3 cells cultured in another naive condition termed PXGL[21] (Methods, Extended Data Fig. 1g, h). We found that the HT method was more robust than the TH method for generating blastoids.

We tested different parameters and found that starting with around 20–30 cells resulted in the highest blastoid formation efficiency (Extended Data Fig. 2a, b), and that mixing TSM and 5i/L/A at 2:1, 1:1 and 1:2 ratios resulted in comparable blastoid formation efficiencies, whereas culturing cells in TSM or 5i/L/A alone generated few or no cavity-containing structures (Extended Data Fig. 1d). We also attempted to generate blastoids in the absence of individual TDM components and found that the inhibitors A83-01 and PD0325901 were most critical, with beneficial effects also seen using WH-4-023 and SB590885. Treatment with Y27632 greatly compromised blastoid formation efficiency (Extended Data Fig. 2c).

The dimensions and cellular composition of human blastoids were comparable to those of human blastocysts at embryonic day 6 (E6) (Extended Data Fig. 1i, j). Through embryo grading, blastoids generated by the HT method exhibited higher ICM but lower trophectoderm ratings compared to human blastocysts, whereas both ICM and trophectoderm ratings in blastoids generated by the TH method were lower than those of human blastocysts (Extended Data Fig. 1k). In summary, we developed a 3D sequential differentiation strategy that enables naive human PSCs to self-organize into blastocyst-like structures in vitro.

Lineage composition of human blastoids

Next, we determined the lineage composition of human blastoids by immunofluorescence analysis. We found that outer cells (TLCs) stained positive for several trophectoderm markers, and that OCT4 was highly expressed in ICM-like cells but maintained a low intensity in outer TLCs (Fig. 1e, Extended Data Fig. 3a–d, h). ICM-like cells expressed either SOX2 or GATA6—marking ELCs or HLCs, respectively—whereas TLCs only weakly expressed GATA6 (Fig. 1f, Extended Data Fig. 3h). These results are consistent with those reported for human E6–E7 blastocysts[26]. In addition, we found that most GATA6^{high} HLCs co-expressed SOX17, and ELCs expressed naive pluripotency markers (KLF4, KLF17, TFCP2L1, SUSD2) but not markers characteristic of formative or primed pluripotency (CD24, OTX2) or primitive streak (EOMES, T) (Extended Data Fig. 3d–g). Triple co-staining revealed similar expression patterns of SOX2, SOX17 and GATA3 in blastocysts and blastoids (Fig. 1g, h, Extended Data Fig. 3i). We also quantified the numbers of SOX2$^+$, SOX17$^+$ and GATA3$^+$ cells in triple-positive blastocysts and blastoids. Compared to blastocysts, blastoids contained more SOX2$^+$ cells and fewer SOX17$^+$ and GATA3$^+$ cells (Extended Data Fig. 3j, k). Notably, we observed SOX2$^+$SOX17$^+$, SOX17$^+$GATA3$^+$ and/or SOX2$^+$GATA3$^+$ cells in both blastoids and blastocysts (Extended Data Fig. 3k).

We co-stained all aggregates using OCT4 and GATA3 antibodies and found that: (1) 85.5% (HT) and 84.3% (TH) of the structures lacked a cavity but contained both OCT4$^+$ cells (mostly inside) and GATA3$^+$ cells (mostly outside); (2) 1.7% (HT) and 6.3% (TH) of the structures resembled trophoblast spheroids; and (3) 12.8% (HT) and 9.4% (TH) of structures not only had a cavity, but also contained inner OCT4$^+$ cells and outer GATA3$^+$ cells (Extended Data Fig. 3l, m).

To gain insights into the dynamics of cell-lineage differentiation during blastoid formation, we performed a time-course immunofluorescence analysis of aggregates generated by the HT method. We noted that GATA6$^+$ and GATA3$^+$ cells first appeared on days 3 and 5, respectively. Between days 7 and 9, GATA6$^+$, GATA3$^+$ and SOX2$^+$ cells segregated (Extended Data Fig. 4a). Blastoids continued to increase in size and cell number after two more days of culture in TDM, although

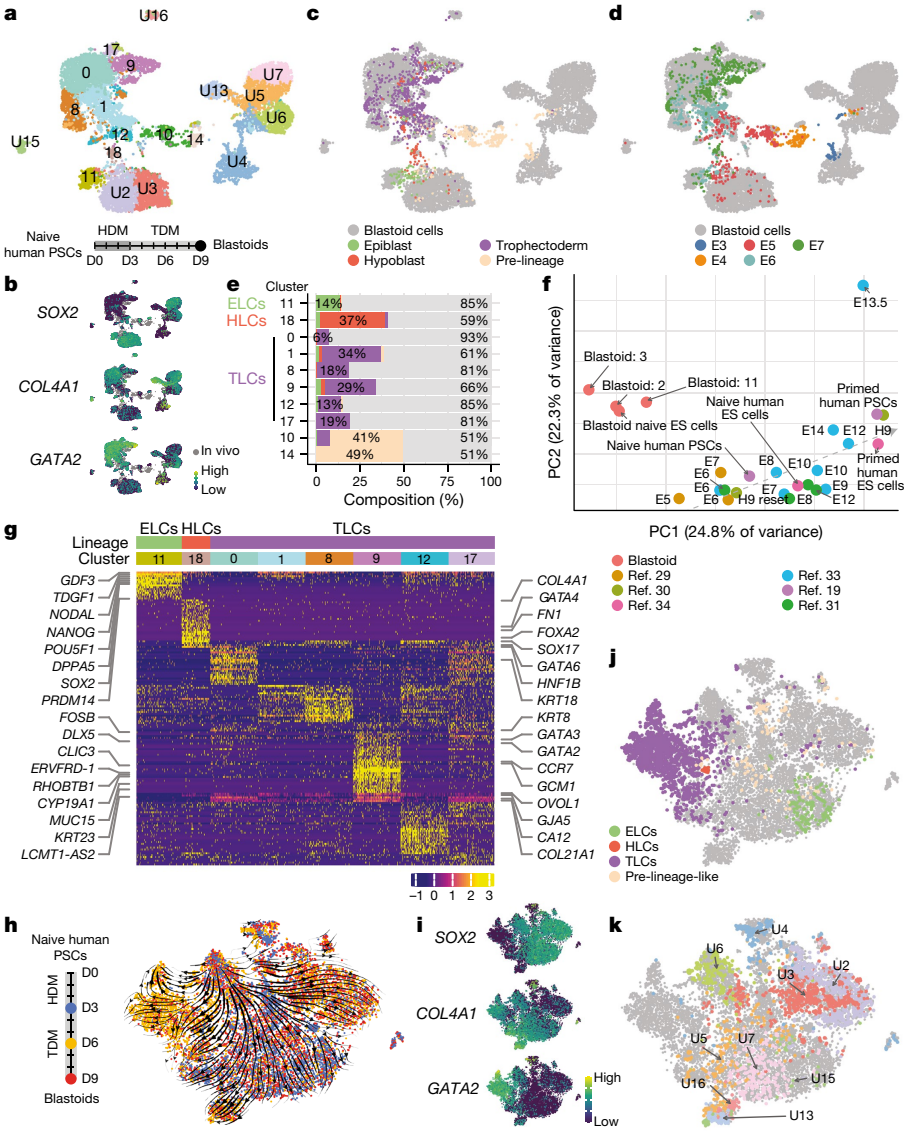

Fig. 2 | Blastoid cell transcriptomes globally resemble those of human pre-implantation embryos. a, Joint uniform manifold approximation and projection (UMAP) embedding of single-cell transcriptomes ($n = 11,182$ cells) from blastoids ($n = 2$ batches of blastoid preparations) and human pre-implantation embryos. U indicates unclassified clusters. D, day. **b**, Gene expression of three lineage markers: *SOX2* (epiblast), *COL4A1* (hypoblast) and *GATA2* (trophectoderm). Grey indicates cells derived from human pre-implantation embryos. **c**, Cells derived from pre-implantation embryos in **a** are coloured by origin: epiblast (green), hypoblast (red) and trophectoderm (purple). Grey indicates blastoid-derived cells. **d**, Cells derived from pre-implantation embryos in **a** are coloured by developmental time: E3–E7. Grey indicates blastoid-derived cells. **e**, Percentage of cells in each cluster derived from blastoids or human pre-implantation embryos. **f**, Principal component analysis (PCA) of blastoid-derived ELCs relative to published maps from human embryos and established human PSC lines[19,29–31,33,34]. Embryonic time points are indicated (labelled as in each previous study). The dotted grey line indicates a developmental time axis estimated from human embryo datasets. **g**, Heat map indicating the expression of lineage-specific genes in blastoid cells. For visualization purposes, clusters with more than 100 cells are randomly down-sampled to 100. In **f**, **g**, clusters contain only blastoid cells, and do not contain cells from human embryos. **h**–**k**, scRNA-seq was performed at days 3, 6 and 9 during blastoid derivation (HT method) ($n = 3$). **h**, *t*-distributed stochastic neighbour embedding (t-SNE) of sequenced cells ($n = 14,191$). Arrows indicate cell trajectories inferred by RNA velocity analysis (right). **i**, Expression of three lineage markers: *SOX2*, *COL4A1* and *GATA2*. **j**, **k**, Embryo-like clusters (**j**) and unclassified clusters (**k**) defined in **a**.

the proportion of proliferating (EdU^+ or $Ki67^+$) cells gradually decreased (Extended Data Fig. 4b–d).

Together, our results demonstrate the coexistence of cells resembling epiblast, trophectoderm and hypoblast lineages in human blastoids, and show that these cells can adopt a spatial organization similar to that of human blastocysts.

Single-cell transcriptome profiling

To determine the transcriptional states of human blastoid cells, we performed single-cell RNA sequencing (scRNA-seq) (Supplementary Table 1). Cell clusters[27,28] exhibited specific expression of marker genes of different lineages, including *SOX2* (epiblast), *COL4A1* (hypoblast) and *GATA2* (trophectoderm), among others (Fig. 2a, b, Extended Data Fig. 5a, d, e). Blastoids sequenced from three independent biological replicates, in which two different starting culture conditions for the naive human PSCs (5i/L/A and PXGL) and two different methods of generation (TH and HT) were used, showed similar results (Extended Data Fig. 5b, c), indicating that blastoid derivation is reproducible.

Next, we performed a comparative analysis with published single-cell transcriptomes derived from human pre-implantation embryos[29]. Blastoid-derived cells clustered with embryo-derived cells from all

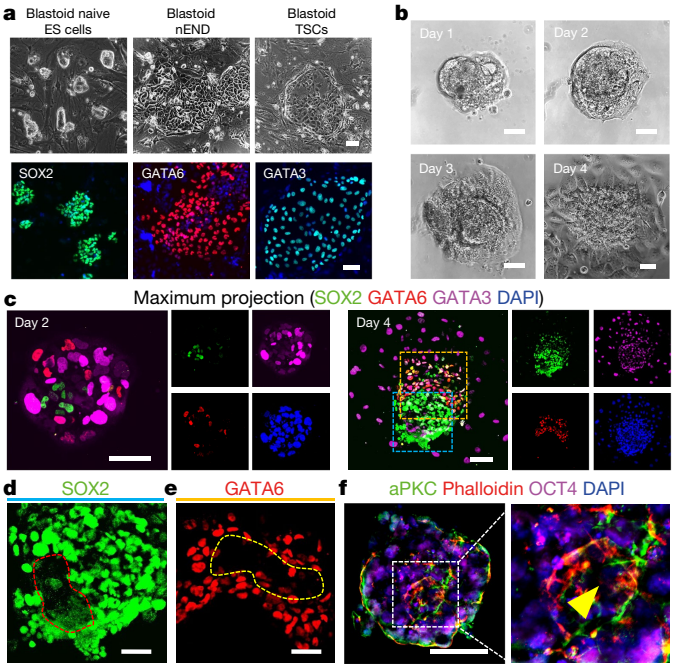

Fig. 3 | Functional assessment of human blastoids. a, Representative phase-contrast (top) and immunofluorescence images (bottom) of blastoid naive ES cells, blastoid nEND and blastoid TSCs. Scale bars, 100 µm. **b**, Representative phase-contrast images of IVC blastoids at days 1–4. Scale bars, 100 µm. **c**, Representative immunofluorescence co-staining images of SOX2, GATA6, and GATA3 in IVC blastoids at day 2 (left) and day 4 (right). Scale bars, 100 µm. **d**, **e**, Higher-magnification images of the boxed areas in **c**. Red dashed line encircles an amniotic-cavity-like structure (**d**); yellow dashed line encircles a yolk-sac-like structure (**e**). Scale bars, 50 µm. **f**, Representative immunofluorescence co-staining image of aPKC, phalloidin and OCT4 in an IVC blastoid at day 4. A higher-magnification image of the boxed area is shown on the right. Yellow arrowhead indicates the amniotic-cavity-like structure. Scale bar, 100 µm. Images in **a**–**f** are representative of at least three independent experiments.

three lineages across developmental time (Fig. 2c, d). We used this overlap to identify and annotate blastoid cells that most closely resemble embryonic cell states (Fig. 2e). Cluster 11 represents ELCs and consists of 40% ($P < 2.2 \times 10^{-16}$, Fisher's test; Supplementary Table 1) of all embryo-derived epiblast cells. Similarly, cluster 18 represents HLCs ($P < 2.2 \times 10^{-16}$, Fisher's test). Six clusters (0, 1, 8, 9, 12, 17) represent TLCs, and contain 95% of all embryo-derived trophectoderm cells ($P < 2.2 \times 10^{-16}$, Fisher's test). Notably, a subset of blastoid cells also clustered with cells from E3–E4 human embryos that have not yet committed to epiblast, hypoblast or trophectoderm lineages (denoted as pre-lineage) (Fig. 2c, d). To assess the temporal identity of blastoid cells, we compared our data with published datasets of in-vitro-cultured human pre- and post-implantation embryos and established stem cell lines[19,20,29–34] (Fig. 2f, Extended Data Fig. 5g). By projecting blastoid cells onto an axis of developmental time generated from embryo-derived cells, we found that blastoid ELCs, HLCs and TLCs most resemble cells from pre-implantation human blastocysts. When compared with the dataset from in vivo cynomolgus monkey embryos[35], blastoid ELCs, HLCs and TLCs correlate most closely with monkey pre-epiblast, hypoblast and trophectoderm cells, respectively (Extended Data Fig. 6a).

Each of the blastoid lineages exhibited distinct patterns of gene expression[29,36] (Fig. 2g, Extended Data Fig. 5d–f) that are consistent with their corresponding lineage in pre-implantation human embryos. For example, epiblast markers (SOX2, NODAL, NANOG) are specific to cluster 11, hypoblast markers (FOXA2, GATA4, COL4A1) are specific to cluster 18 and trophectoderm markers (GATA2, GATA3) are most highly expressed in trophectoderm-like clusters. Notably, cluster 9 expresses polar trophectoderm markers[29] (CYP19A1, MUC15, OVOL1), suggesting a polar-trophectoderm-like identity (Fig. 2g, Extended Data Fig. 5d–f). In agreement, CCR7, a polar trophectoderm marker[29], is most highly expressed in blastoid TLCs adjacent to ICM-like cells (Extended Data Fig. 6h). We also performed a gene ontology analysis, which showed a consistent lineage-specific enrichment of biological processes (Extended Data Fig. 5h, Supplementary Table 2), with ELCs enriched for stem cell population maintenance, TLCs for placenta development ($P = 4.5 \times 10^{-11}$) and HLCs for embryonic morphogenesis ($P = 3.9 \times 10^{-7}$).

To examine the temporal steps of blastoid derivation, we performed scRNA-seq across three time points (HT method, days 3, 6 and 9) (Fig. 2h, Extended Data Fig. 6i). By the end of day 3 using HDM culture, ELCs ($SOX2^+$) and HLCs ($COL4A1^+$) are well-represented (Fig. 2i, Extended Data Fig. 6b–d). By contrast, TLCs ($GATA2^+$) appear only after TDM culture (days 6 and 9). Cell-state trajectory analysis suggests that ELCs, HLCs and TLCs may be derived from an unclassified population of cells (cluster U6; Fig. 2h, j). Notably, cluster U6 contains a small number of cells from early (E3–E4) human embryos and exhibits expression of markers of multiple lineages, suggesting that these U6 cells represent an early transitional cell state (Fig. 2d). Consistent with our cell-state trajectory analysis, pseudotime analysis[37] places blastoid ELCs on a trajectory between U6 and epiblast cells (Extended Data Fig. 6e). This cell-state transition involves the activation of genes that are expressed during early epiblast development in pre-implantation human embryos[31], such as *WNT5B* (Extended Data Fig. 6f, g). By contrast, these genes are not activated in $SOX2^+$ cells in unclassified clusters U2 and U3 ($P = 1.5 \times 10^{-34}$, binomial test) (Supplementary Table 1), and which instead exhibit activation of trophectoderm-associated genes including *GATA3* ($P = 5.1 \times 10^{-48}$; Extended Data Fig. 6d). These results suggest that clusters U2 and U3 may represent cells that respond to TDM growth conditions but that have not properly established blastocyst-like states. In addition, clusters U5, U7, U13, U15 and U16 express hypoblast marker genes and transcriptionally cluster near HLCs (Fig. 2k). Finally, lending in vivo support, some unclassified clusters contain a small number of cells from human pre-implantation embryos.

Derivation of stem cell lines

Using 5i/L/A[22], TSM[23] and NACL[20] (Methods) conditions, we successfully derived naive ES cells, trophoblast stem cells (TSCs) and naive extra-embryonic endoderm lines from individually plated blastoids (denoted as blastoid naive ES cells, blastoid TSCs and blastoid nEND, respectively) (Fig. 3a, Extended Data Fig. 7a–d). scRNA-seq analysis showed that blastoid naive ES cells, nEND and TSCs segregated into distinct clusters, and expressed SOX2, GATA6 and GATA3, respectively (Extended Data Fig. 7e). Comparison with published datasets[19,20,30,34] provided additional support for their lineage identities (Extended Data Fig. 7f). We also confirmed that blastoid TSCs and blastoid nEND can be directly derived from TLCs and HLCs, respectively, rather than converted from ELCs (Extended Data Fig. 7g–l).

To ascertain the identities and developmental potency of blastoid-derived stem cells, we conducted in vitro differentiation assays. Following an established protocol[23], we differentiated blastoid TSCs into extravillous trophoblast (EVT)- and syncytiotrophoblast (SCT)-like cells. These lineage identities were determined by two lines of evidence. First, immunofluorescence analyses showed that EVT- and SCT-like cells expressed the EVT marker HLA-G and the SCT marker SDC1, respectively, stained positive for cytotrophoblast markers GATA3, CK7 and CGB, and lost the expression of the TSC marker CDH1. Second, quantitative PCR with reverse transcription (RT–qPCR) revealed that several reported markers of EVTs and SCTs were upregulated in EVT- and SCT-like cells when compared to blastoid TSCs, respectively (Extended Data

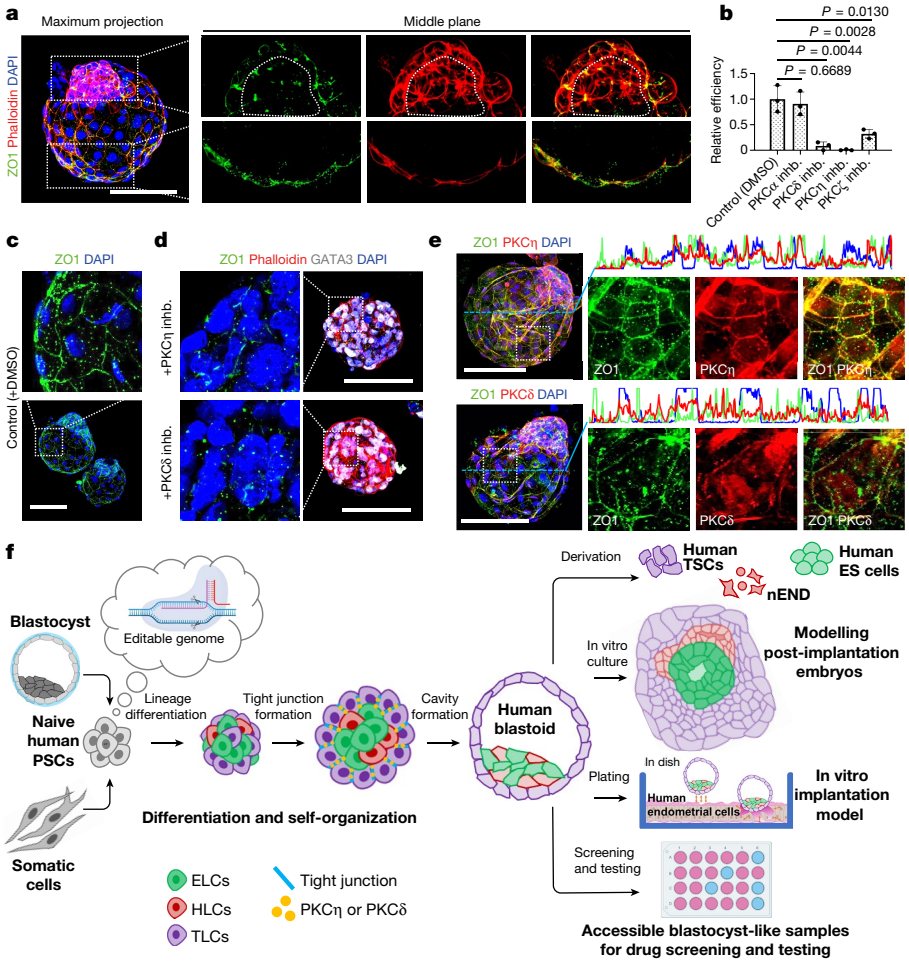

Fig. 4 | PKC isozymes regulate cavity formation in human blastoids. **a**, Representative immunofluorescence co-staining images of ZO1 and phalloidin in a human blastoid. Higher-magnification images of the boxed areas are shown on the right. White dotted lines encircle the ICM-like compartment. Scale bar, 100 μm. **b**, Relative blastoid formation efficiency after treatments with different PKC inhibitors ($n = 3$ biological replicates; mean ± s.d.; two-tailed t-test). **c**, Representative immunofluorescence images of ZO1 in control dimethyl sulfoxide (DMSO)-treated human blastoids. A higher-magnification image of the boxed area is shown at the top. Scale bar, 100 μm. **d**, Representative immunofluorescence co-staining images of ZO1, phalloidin and GATA3 in cell aggregates after treatments with inhibitors of PKCη (top) and PKCδ (bottom). Higher-magnification images of ZO1 in the boxed areas are shown on the left. Scale bars, 100 μm. **e**, Representative immunofluorescence images of ZO1 and PKCη (top), or ZO1 and PKCδ (bottom) in human blastoids. Top right, line intensity profiles of the blue dashed lines; bottom right, higher-magnification images of the boxed areas. Scale bars, 100 μm. **f**, Schematic summary of human blastoid formation from naive human PSCs and their potential applications. Images in **a**, **c**–**e** are representative of at least three independent experiments.

Fig. 8a–c). For blastoid nEND, we performed random differentiation and generated extra-embryonic-mesenchyme-cell-like (EXMC-like, COL6A1$^+$GATA6^{low}FOXA1^{low})[35,38] and visceral-and yolk-sac-endoderm-like (VE/YE-like, FOXA1$^+$GATA6$^-$COL6A1$^-$) cells (Extended Data Fig. 8d–f). RT–qPCR analysis confirmed the upregulation of several EXMC and VE/YE markers in differentiated cultures when compared to blastoid nEND (Extended Data Fig. 8g). Blastoid naive ES cells could be converted to primed human ES cells capable of generating tissues from all three germ layers in teratomas (Extended Data Fig. 8h–j).

To further validate lineage identities, we performed microinjections of blastoid-derived stem cells into mouse blastocysts followed by in vitro embryo culture[39]. After 5–6 days of culture, 23.9%, 8.0% and 15.8% of embryos contained derivatives of TSCs (GATA3$^+$), nEND (GATA6$^+$) and naive ES cells (SOX2$^+$) in their corresponding lineages, respectively (Extended Data Fig. 7m–o, Supplementary Table 3).

Together, these results demonstrate that human blastoids are amenable to stem cell derivation from all blastocyst lineages, and that these blastoid-derived stem cells can further differentiate into respective downstream cell types.

In vitro culture of human blastoids

To determine whether human blastoids can self-organize into structures similar to peri-implantation embryos[26,40], we cultured them for an additional four days in in vitro culture (IVC) media[41]. Around 40–50% of blastoids attached to the culture plate after 1–2 days and adopted a flattened structure similar to that of human embryos grown in IVC media[26,41] (Fig. 3b). At IVC day 2, clearly segregated lineages were observed in some outgrowths, with SOX2$^+$ cells situated in the centre surrounded by GATA6$^+$ and GATA3$^+$ cells (Fig. 3c). The segregation of SOX2$^+$ and GATA6$^+$ cells in human blastoids is consistent with IVC-cultured human embryos at E8 and contrasts with mouse embryos, in which the segregation of epiblasts and hypoblasts occurs before implantation[26,41]. At IVC day 4, around 10% of outgrowths contained segregated SOX2$^+$, GATA6$^+$, and GATA3$^+$ cells. Among these outgrowths, some exhibited abnormal growth (Extended Data Fig. 9a), whereas others showed features resembling IVC-cultured human embryos, as evidenced by several observations: (1) the formation of an amniotic-cavity-like structure from ELCs (SOX2$^+$OCT4$^+$), which gained OTX2 expression

(Fig. 3c, d, f, Extended Data Fig. 9b, c); (2) the preferential localization of HLCs ($GATA6^+$) to one side of the ELCs, and the formation of a yolk-sac-like cavity (Fig. 3c, e); and (3) the cytoplasmic filamentous staining pattern of cytokeratin (CK7), the expression and secretion of human chorionic gonadotrophin β (HCGB), and evidence of multinucleation in outer cells—consistent with their trophoblast-like, rather than amnion-like, identity (Extended Data Fig. 9d–f). Thus, human blastoids can—albeit at low frequency—self-organize into structures with features of peri-implantation human embryos[26,40].

Cavity formation in human blastoids

The generation and maintenance of the blastocoel is critical for proper blastocyst function. Blastocoel formation depends on the assembly and maturation of tight junctions to help seal trophectoderm cells and enable fluid accumulation and blastocyst expansion. Although the role of tight junctions in blastocoel formation has been extensively studied in mice, limited information is available in humans. We performed a time-course immunofluorescence analysis of cell aggregates during blastoid formation using antibodies against phalloidin and zonula occludens-1 (ZO1). Continuous ZO1 staining could be observed in outer cells beginning at day 5 and was maintained in the TLCs of formed blastoids. By contrast, ICM-like cells exhibited punctate ZO1 staining, consistent with a non-epithelial phenotype (Fig. 4a, Extended Data Fig. 10a). Notably, human blastoids underwent cavity expansion[42], showing that functional tight junctions are present in TLCs (Supplementary Video 1).

The protein kinase C (PKC) family regulates the assembly, maintenance and function of tight junctions, and thereby affects epithelial barriers that are important for blastocoel formation. A pan-PKC inhibitor (Gö6983) prevented cavity formation in human blastoids, whereas an inhibitor selective for the PKCα and PKCβ isozymes (Gö6976) had minimal effect (Extended Data Fig. 10b). These results show that PKC signalling has a critical role in the formation of human blastoids, and that its action depends on specific isozyme(s). To identify the PKC isozyme(s) involved, we used isozyme-specific peptide inhibitors that target PKCδ, PKCζ and PKCη, which are expressed in human embryos and implicated in tight junction assembly in mouse blastocysts[43,44] (Extended Data Fig. 10c). Inhibiting PKCδ and PKCη—and, to a lesser extent, PKCζ—prevented cavity formation in human blastoids (Fig. 4b). Consistent with this finding, there was little continuous ZO1 staining between $GATA3^+$ cells after inhibition of the δ, ζ and η isozymes of PKC (Fig. 4c, d, Extended Data Fig. 10d, e). In addition, we found that PKCδ and PKCη localized to the junctions between TLCs and overlapped with ZO1 staining (Fig. 4e). Thus, specific PKC isozymes are involved in the assembly and/or maintenance of tight junctions in TLCs and are essential for cavity formation in human blastoids. Additional studies are required to confirm the function of these PKC isozymes during blastocoel formation in human blastocysts.

Discussion

In summary, by leveraging the embryonic and extra-embryonic bipotency of naive human PSCs, we have developed a strategy that enables the generation of human-blastocyst-like structures through sequential lineage differentiation and self-organization (Fig. 4f). Human blastoids resemble human blastocysts in morphology, cell-lineage composition and allocation, and transcriptional state. They also recapitulate key morphogenetic events that occur during human peri-implantation development.

Our present blastoid model has several limitations: (1) the derivation efficiency varies between batches, cell lines, starting conditions and derivation methods; (2) the derivation process is lengthy and unsynchronized; (3) some blastoid cell states do not match human blastocysts; and (4) the efficiency of generating early post-implantation structures is low. These drawbacks are potentially attributable to the suboptimal existing naive human PSC culture conditions, and the two-step derivation protocols that deviate from the lineage specifications that are observed in human blastocysts[29]. Studies are needed to further improve the human blastoid model to produce a more faithful substitute for human blastocysts. Notably, whereas blastoids can model certain aspects of early human development, evidence from mouse studies suggests that they cannot generate viable embryos[16–18].

Nonetheless, human blastoids provide a valuable model for both basic and translational research as an accessible alternative to human blastocysts. Blastoid generation is scalable and amenable to well-controlled perturbation experiments, and will offer a unique model to study human embryogenesis, early developmental defects, early pregnancy loss and new contraceptives (Fig. 4f).

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-021-03356-y.

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Methods

Data reporting

No statistical methods were used to predetermine sample size. The experiments were not randomized and the investigators were not blinded to allocation during experiments and outcome assessment.

Ethics statement

All human ES cell, iPS cell and blastoid experiments were performed at the UT Southwestern Medical Center and followed the 2016 Guidelines for Stem Cell Research and Clinical Translation released by the International Society for Stem Cell Research (ISSCR). Human ES cell, iPS cell and blastoid work was reviewed and approved by the UT Southwestern Stem Cell Oversight Committee (SCRO) (registration no. 29).

All human-embryo-related experiments were conducted at Kunming Medical University. The studies involving human embryos were approved by the Medical Ethical Committee of Kunming Medical University. Informed consent was obtained from all couples donating the blastocysts. The embryos were donated by couples undergoing in vitro fertilization (IVF); all donated embryos were in excess of clinical need and all couples had at least one healthy baby after IVF clinic treatment. The informed consent process for embryo donation complies with the ISSCR Guidelines for Stem Cell Research and Clinical Translation (2016) and the Ethical Guidelines for Human Embryonic Stem Cell Research (2003) jointly issued by the Ministry of Science and Technology and the Ministry of Health of People's Republic of China. No financial incentives were offered for the donations. In the process, couples were informed that their embryos would be used to study the developmental mechanisms of human embryos and that their donation would not affect their in vitro fertilization cycle.

Thawing and culture of human embryos

All donated human embryos were cryopreserved using a vitrification kit (KITAZATO BioPharma). In brief, the embryos were transferred into equilibration solution for 8 min at room temperature. Then the embryos were loaded into a Cryotop (Kitazato BioPharma) in vitrification solution and transferred to liquid nitrogen within 1 min. Vitrified embryos were thawed using a thawing kit (Kitazato BioPharma). In brief, embryos were thawed by placing the Cryotop in a 37 °C pre-warmed thawing solution; then collected, transferred and incubated in diluent solution for 3 min at room temperature; and then transferred and incubated in washing solution 1 for 5 min and washing solution 2 for 5 min at room temperature. Finally, the thawed embryos were recovered in a 37 °C and 6% CO_2 incubator for 2 h. Human embryo culture was stopped at E6 and no primitive streak formed.

Mice

CD-1 (ICR) mice were purchased from Charles River or Envigo. NOD-SCID (NOD.CB17-*Prkdc*^{scid}/J) mice were purchased from The Jackson Laboratory. Mice were housed in a 12-h light–12-h dark cycle at 22.1–22.3 °C and 33–44% humidity. All procedures related to animals were performed in accordance with the ethical guidelines of the University of Texas Southwestern Medical Center. The animal protocol was reviewed and approved by the UT Southwestern Institutional Animal Care and Use Committee (IACUC) (protocol no. 2018-102430). All experiments followed the ISSCR Guidelines for Stem Cell Research and Clinical Translation (2016). Ex vivo human–mouse chimeric studies were reviewed and approved by the UT Southwestern SCRO (registration no. 14).

Culturing of naive human PSCs

Naive WIBR3 human ES cells[22] were obtained from R. Jaenisch and T. Theunissen. Primed HFF human iPS cells and BJ human iPS cells were generated using episomal vectors from human foreskin fibroblast (HFFs; ATCC CRL-2429) and BJ fibroblasts (ATCC CRL-2522), respectively, which were reported previously[24,25]. Naive HFF human iPS cells and BJ human iPS cells were converted from primed HFF human iPS cells[24] and BJ human iPS cells[25] following a previously described protocol[22]. All naive human PSCs were cultured on mitotically inactivated mouse embryonic fibroblast (MEF) feeders as described previously[19,21]. In brief, around 2×10^5 cells were plated into 6-well plates pre-coated with MEFs in 5i/L/A or PXGL medium supplemented with 5 μM Y-27632 (Selleckchem). The cells were passaged with TrypLE Express (Gibco) once they reached around 70–80% confluency (usually every 5–6 days). The 5i/L/A medium (500 ml) was prepared using the following: 250 ml DMEM/F12 (Invitrogen), 250 ml neurobasal medium (Invitrogen), 5 ml N2 supplement (Invitrogen), 10 ml B27 supplement (Invitrogen), 1× GlutaMAX (Gibco), 1× nonessential amino acids (Gibco), 0.1 mM β-mercaptoethanol (Gibco), 0.5% penicillin–streptomycin (Gibco), 50 mg ml^{-1} bovine serum albumin (BSA, Sigma) and the following small molecules and cytokines: 1 μM PD0325901 (Stemgent), 0.5 or 1 μM IM-12 (Enzo), 0.5 μM SB590885 (R&D systems), 1 μM WH-4-023 (A Chemtek), 20 ng ml^{-1} recombinant human LIF (Peprotech) and 10 ng ml^{-1} activin A (Peprotech). PXGL medium (500 ml) was prepared using the following: 250 ml DMEM/F12, 250 ml neurobasal medium (Invitrogen), 5 ml N2 supplement, 10 ml B27 supplement, 1× GlutaMAX, 1× nonessential amino acids, 0.1 mM β-mercaptoethanol, 0.5% penicillin–streptomycin and the following small molecules and cytokines: 1 μM PD0325901, 2 μM XAV939 (Sigma), 2 μM Gö6983 (Sigma) and 20 ng ml^{-1} recombinant human LIF.

Monolayer differentiation of naive human PSCs

Naive human PSCs were dissociated into single cells by incubation with TrypLE Express at 37 °C for 3 min. Cells were collected and resuspended in naive human PSC medium after centrifugation at 200*g* for 5 min. Then, the cell suspension was incubated in a gelatinized tissue culture plate for 30 min at 37 °C in 5% CO_2 to remove MEF cells. The supernatant containing naive human PSCs was collected and replated into a 12-well plate (100,000 cells per well) pre-coated with Matrigel (BD Biosciences) in naive human PSC medium supplied with 5 μM Y-27632 (Reagents Direct). For TLC and HLC differentiation, after 2 days, the naive human PSC medium was replaced with TSM or HDM and cells were cultured for an additional seven days. For TLC and HLC codifferentiation, the medium was sequentially replaced with HDM for three days, then TSM for five days, or with TSM for five days, then HDM for three days, with medium changed daily. The TSM was prepared by inclusion of the following: DMEM/F12 supplemented with 0.1 mM β-mercaptoethanol, 0.2% fetal bovine serum (FBS, Gibco), 0.5% penicillin–streptomycin, 0.3% BSA, 1% ITS-X (Gibco), 1.5 μg ml^{-1} L-ascorbic acid (Wako), 50 ng ml^{-1} EGF (Peprotech), 2 μM CHIR99021 (Selleckchem), 0.5 μM A83-01 (Sigma), 1 μM SB431542 (Sigma) and 0.8 mM VPA (Sigma). The HDM was prepared using the following: 1:1 (v/v) mixture of DMEM/F12 and neurobasal medium, 1× N2 supplement, 1× B27 supplement, 1× GlutaMAX, 1× nonessential amino acids, 0.1 mM β-mercaptoethanol, 0.5% penicillin–streptomycin, 20 ng ml^{-1} bFGF (Peprotech), 20 ng ml^{-1} activin A and 3 μM CHIR99021.

Generation of human blastoids

The quality of starting naive human PSCs is critical for human blastoid generation. We recommend using stable naive human PSCs at an early passage (lower than 20 passages). For the generation of human blastoids, 60–70% confluent naive human PSCs were dissociated into single cells by incubation with TrypLE Express at 37 °C for 3 min. Cells were collected and resuspended in naive human PSC medium after centrifugation at 200*g* for 5 min. Then, the cell suspension was incubated in a gelatinized tissue culture plate for 30 min at 37 °C in 5% CO_2 to remove MEF cells. The supernatant containing naive human PSCs was collected and passed through a 40-μm cell strainer, and cells were counted. Meanwhile, the AggreWell 400 (STEMCELL Technologies) was prepared according to the manufacturer's instructions. In brief, wells were rinsed with anti-adhesion rinsing solution (STEMCELL

Technologies), centrifuged at 2,000*g* for 5 min and incubated at room temperature for 10 min. After incubation, wells were washed with naive human PSC medium once and 0.5 ml fresh naive human PSC medium containing 5 μM Y-27632 (Selleckchem) was added.

Approximately 30,000 cells (around 25 cells per microwell for 5i/L/A-cultured WIBR3 human ES cells; for other cell lines the starting number needs to be optimized) were resuspended in 1 ml 5i/L/A medium supplemented with 5 μM Y-27632 (Selleckchem) and seeded into one well of a prepared AggreWell 400 24-well plate. The plate was centrifuged at 200*g* for 1 min and placed in a 37 °C incubator. Aggregates formed after around 12 h of culture, and the medium was replaced with HDM or TDM. The day of switching from naive human PSC medium to differentiation medium (HDM or TDM) was designated as day 1.

For the HT method, naive human PSC medium is first changed to HDM (day 1). On day 3, carefully remove as much HDM as possible, and then add 1 ml of TDM. Repeat this step one additional time to help completely remove the remaining HDM. Fresh TDM is replaced every two days. After six to eight days of culture in TDM (usually seven days for 5i/L/A-cultured WIBR3 cells, but the time may vary between different cell lines, different starting naive human PSC culture conditions, different starting cell number, and different batches), human blastoids can be observed in some microwells. For the TH method, naive human PSC medium is first changed to TDM (day 1). Fresh TDM is replaced every two days until the cavity structures appear (usually it takes six to seven days for 5i/L/A-cultured WIBR3 cells, but the time may vary as mentioned above), then the medium is replaced with HDM and cultured for one to two days. One or two days (using the TH method) and two or three days (using the HT method) of HDM treatment showed no substantial difference in blastoid formation efficiency. In our experience, the HT method was more robust than the TH method. We observed that the TH method is more sensitive to the quality and passage number of starting naive human PSCs than the HT method.

All formed blastoids were manually isolated using a mouth pipette under a stereomicroscope for downstream experiments. The TDM was prepared using the following: 1:1 (v/v) mixture of DMEM/F12 and neurobasal medium, 0.5× N2 supplement, 0.5× B27 supplement, 0.5% ITS-X, 0.5× GlutaMAX, 0.5× nonessential amino acids, 0.1 mM β-mercaptoethanol, 0.5% knockout serum replacement (KSR, Gibco), 0.1% FBS, 50 mg ml^{-1} BSA, 0.5% penicillin–streptomycin, 1 μM PD0325901, 0.5 μM A83-01, 0.25 μM SB590885, 0.5 μM WH-4-023, 0.25 μM IM-12, 1 μM CHIR99021, 0.5 μM SB431542, 10 ng ml^{-1} recombinant human LIF, 25 ng ml^{-1} EGF, 0.75 μg ml^{-1} L-ascorbic acid and 0.4 mM VPA. For certain experiments the following chemicals were added to the TDM: 100 nM Gö6976 (Selleckchem), 1 μM Gö6983 (Selleckchem), 10 μM PKCα (C2-4) inhibitor peptide (Cayman), 10 μM PKCζ pseudosubstrate inhibitor (Cayman), 10 μM PKCη pseudosubstrate inhibitor (Sigma) or 10 μM PKCδ inhibitor KAI-9803 (delcasertib hydrochloride, MedChemExpress). A detailed step-by-step protocol for generating human blastoids from naive human PSCs is available on Protocol Exchange[45].

Quantification of blastoid formation efficiency

Cell aggregates were collected from microwells by gently pipetting up and down 1–2 times with a 1-ml pipette. To minimize the shearing force, around 1 mm from the end of the 1-ml pipette tips were cut off before using. All of the cell aggregates were transferred into a well of a six-well plate and counted under a stereomicroscope. The aggregates with the presence of an ICM-like compartment, a trophectoderm-like compartment and a visible cavity were counted as blastoids. The blastoid formation efficiency was calculated as the number of blastoids per number of total aggregates.

Derivation of stem cell lines from human blastoids

The previously described protocols to derive and culture naive human ES cells[22], human nEND[20] and human TSCs[23] were used. In brief, individual human blastoids were transferred onto MEFs and cultured in 5i/L/A medium (for blastoid naive ES cells), NACL medium (for blastoid nEND) and human TSC medium (for blastoid TSCs). Within four to five days, outgrowths could be observed. Outgrowths were dissociated with TrypLE Express and passaged onto newly prepared MEF plates. Individual colonies were manually picked, dissociated and cultured in their corresponding media. NACL medium was prepared using the following: 1:1 (v/v) mixture of DMEM/F12, and neurobasal medium, 1× N2 supplement, 1× B27 supplement, 1× GlutaMAX, 1× nonessential amino acids, 0.1 mM β-mercaptoethanol, 0.5% penicillin–streptomycin, 100 ng ml^{-1} activin A, 3 μM CHIR99021 and 10 ng ml^{-1} recombinant human LIF.

For certain experiments, (1) the trophectoderm-like layer of blastoids was manually isolated using a microblade and plated onto MEFs in human TSC medium (for blastoid TSC derivation); or (2) the ELCs (GFP^{+}SUSD2^{+} double-positive cells[46]) were removed from OCT4 reporter (OCT4-GFP) WIBR3[22] human ES cell-derived blastoids by fluorescence-activated cell sorting (FACS), and the resulting GFP^{-}SUSD2^{-} double-negative cells were plated onto MEFs in human TSC medium (for blastoid TSC derivation) or NACL medium (for blastoid nEND derivation).

In vitro differentiation of blastoid-derived stem cell lines

Blastoid TSCs were differentiated as previously reported[23], with modifications. In brief, blastoid TSCs were dissociated into single cells with TrypLE Express and seeded into a six-well plate pre-coated with Matrigel at a density of 1.5×10^5 cells per well. To generate EVT-like cells, cells were cultured in 2 ml of EVT medium supplemented with 2% Matrigel. After three days of culture, the medium was replaced with EVT medium supplemented with 0.5% Matrigel but lacking NRG1, and the cells were cultured for an additional three days. For differentiation of SCT-like cells, 2 ml of SCT medium was added and the cells were cultured for six days, with the medium changed every other day. The EVT medium was prepared using the following: DMEM/F12 supplemented with 0.1 mM β-mercaptoethanol, 0.5% penicillin–streptomycin, 0.3% BSA, 1% ITS-X, 100 ng ml^{-1} NRG1 (Cell Signaling), 7.5 μM A83-01, 2.5 μM Y-27632 and 4% KSR. SCT medium was prepared using the following: DMEM/F12 supplemented with 0.1 mM β-mercaptoethanol, 0.5% penicillin–streptomycin, 0.3% BSA, 1% ITS-X, 2.5 μM forskolin (Sigma), 2.5 μM Y-27632 and 4% KSR.

For blastoid nEND differentiation, cells were dissociated into single cells using TrypLE Express and seeded into a six-well plate pre-coated with 5 μg ml^{-1} laminin (Gibco) at a density of 5×10^5 cells per well in nEND differentiation medium for 9 days, with the medium changed every other day. nEND differentiation medium was prepared using the following: 1:1 (v/v) mixture of DMEM/F12 and neurobasal medium, 1× N2 supplement, 1× B27 supplement, 1× GlutaMAX, 1× nonessential amino acids, 0.1 mM β-mercaptoethanol, 0.5% penicillin–streptomycin and 10% FBS.

Teratoma formation assay

A total of 2×10^6 cells were resuspended in 50 μl DMEM–Matrigel solution (1:1 mixture) and injected subcutaneously into 10-week-old immunodeficient NOD-SCID mice. After six to eight weeks, teratomas were dissected and fixed with Bouin's solution. Paraffin-embedded teratomas were sectioned (10 μm thick) and stained with haematoxylin and eosin (H&E).

Mouse blastocyst injection of blastoid-derived stem cell lines

Single-cell suspensions of monomeric Kusabira-Orange (mKO)-labelled blastoid TSCs, blastoid nEND or blastoid naive ES cells were added to a 40-μl droplet of modified KSOM[47] with 2 mM HEPES containing the blastocysts collected from 8–10-week-old CD-1 mice and placed on an inverted microscope (Nikon) fitted with micromanipulators (Narishige). Individual cells were collected using a micropipette with a 15–20 μm internal diameter, and a Piezo Micro Manipulator (Prime Tech) was

used to create a hole in the zona pellucida and the trophectoderm layer of mouse blastocysts. Ten cells were introduced into the blastocoel. After microinjection, the blastocysts were cultured in modified KSOM with amino acids. Fluorescent signals were observed using an inverted fluorescence microscope (Nikon) at 24 and 48 h post-injection.

In vitro culture of human blastoids

Human blastoids were cultured beyond the implantation stages using a protocol previously described for human blastocysts[40]. In brief, human blastoids were manually isolated using a mouth pipette, washed twice with PBS and transferred into a µ-Slide 8-well (ibidi) containing IVC-1 medium. After 24 h, half of the medium was replaced. Human blastoids usually attached to the plate within two days. Once the blastoids attached, the medium was switched to IVC-2 medium. Half of the IVC-2 medium was replaced every day. IVC-1 and IVC-2 media were prepared using the following: IVC-1: advanced DMEM/F12 (Gibco) supplemented with 20% (v/v) FBS, 2 mM L-glutamine (Gibco), 1× ITS-X, 8 nM β-oestradiol (Sigma), 200 ng ml^{-1} progesterone (Sigma), 25 µM *N*-acetyl-L-cysteine (Sigma), 0.22% (v/v) sodium lactate (Sigma), 1 mM sodium pyruvate (Sigma) and 10 µM Y27632 (Selleckchem); IVC-2: advanced DMEM/F12 supplemented with 30% (v/v) KSR, 2 mM L-glutamine, 1× ITS-X, 8 nM β-oestradiol, 200 ng ml^{-1} progesterone, 25 µM *N*-acetyl-L-cysteine, 0.22% sodium lactate, 1 mM sodium pyruvate and 10 µM Y27632. Of note, all human blastoids were cultured for up to four days in IVC media, which roughly corresponds to around day-10–11 human embryos, and no primitive-streak-like structures formed.

Human blastocyst and blastoid grading

Human blastocysts and blastoids were scored using the Gardner and Schoolcraft (2000) scoring system. By this method, ICMs that were tightly packed containing many cells, loosely grouped with several cells, or with very few cells were scored A, B, or C, respectively. The trophectoderm was scored A if it formed a cohesive epithelium with many cells, B if it formed a loose epithelium with few cells, or C if it contained very few cells.

Immunofluorescence staining

Samples (including: two-dimensional-cultured cells, human blastoids and non-blastoid aggregates) were fixed with 4% paraformaldehyde (PFA) in PBS for 20 min at room temperature and permeabilized with 0.5% Triton X-100 in PBS for 1 h (4 °C overnight for blastoids). Samples were then blocked with blocking buffer (PBS containing 5% FBS, 4% BSA and 0.1% Tween-20) at room temperature for 1 h, or overnight at 4 °C. Primary antibodies diluted in blocking buffer were applied to samples and incubated for 2 h at room temperature or overnight at 4 °C. Samples were washed three times with PBS containing 0.1% Tween-20 (PBS-T), followed by incubation with fluorescently conjugated secondary antibodies diluted in blocking buffer for 2 h at room temperature or overnight at 4 °C. Samples were washed three times with PBS-T. Finally, cells were counterstained with 300 nM 4′,6-diamidino-2-phenylindole (DAPI) solution at room temperature for 20 min. Phalloidin was directly stained along with other secondary antibodies in blocking buffer. For EdU staining, the Click-it EdU Alexa Fluor 488 Imaging Kit (Invitrogen) was used following the manufacturer's instructions.

For human blastocyst staining, whole embryos were fixed in 4% PFA in PBS for 30 min at room temperature. Next, embryos were washed three to five times with PBS, and then stained directly. First, embryos were permeabilized in PBS with 0.3% Triton X-100 for 30 min at room temperature. Then embryos were blocked with 3% (w/v) BSA and 10% (v/v) FBS in PBS overnight at 4 °C. Embryos were then incubated with primary antibodies overnight at 4 °C. After washing, Alexa Fluorophore (488, 594 and 647)-conjugated secondary antibodies were incubated with the slides in the dark at room temperature for at least 2 h.

The antibody information and dilutions used in this study are provided in Supplementary Table 4. Samples were imaged using a fluorescence (Revolve, Echo) or a confocal microscope (A1R, Nikon). The cell numbers ($DAPI^+$, $SOX2^+$, $GATA3^+$, $SOX17^+$, EdU^+ and $Ki67^+$) of blastoids were counted manually using the confocal microscope (A1R, Nikon).

RT–qPCR analysis

Total RNA was isolated using an RNeasy Mini Kit (Qiagen) following the manufacturer's instructions. cDNA was synthesized using iScript Reverse Transcription Supermix kit (Bio-Rad) and amplified with SYBR Green PCR Master Mix (Thermo Fisher Scientific) on a Touch Thermal Cycler Real-Time PCR system (C1000, Bio-Rad). *GAPDH* expression level was used as an internal normalization control. All primers used in this study are listed in Supplementary Table 4.

scRNA-seq library generation

Blastoids were selected for sequencing on the basis of several morphological criteria, including the presence of an ICM-like compartment, a trophectoderm-like compartment and a visible cavity. The same selection criteria were used for all 'end-point' blastoid sequencing experiments across both methods (that is, HT, day 9; TH, day 8) and both starting naive human PSC conditions (5i/L/A, PXGL). Blastoid grading was not used during this selection process. For days 3 and 6 (HT method), we used all aggregates for sequencing. Human blastoids were manually pooled using a mouth pipette, washed three times in PBS and then dissociated with trypsin-EDTA (0.25%) (Gibco) at 37 °C in 5% CO_2 for 40 min with agitation. We directly added 2 ml of PBS containing 0.04% BSA and centrifuged the sample at 500*g* for 5 min. Cell pellets were resuspended in PBS containing 0.04% BSA and then filtered through a 40-µm cell strainer. Cell number was determined using a haemocytometer. Single-cell suspensions (500–1,000 cells per ml) were loaded into a 10X Genomics Chromium Chip following the manufacturer's instructions[48] (10X Genomics, Chromium Next GEM Single Cell 3′ GEM, Library and Gel Bead Kit v.3.1).

Sequencing, base calling and demultiplexing

Binary base call (BCL) files generated by Illumina NextSeq 500/550 and NovaSeq 6000 sequencing systems (Illumina) were demultiplexed and converted to standard FASTQ files using the mkfastq function from the Cell Ranger pipeline (v.4.0.0).

Pre-processing single-cell data

The 10X Genomics single-cell data were processed using the Cell Ranger pipeline (v.4.0.0) with default parameters except the number of cells expected was set to 10,000 to generate the expression matrix. The reference and gene annotations were downloaded from the 10X Genomics website (v.3.0.0, GRCh38). The number of single cells in the assay was determined by the Cell Ranger pipeline. Scrublet was used to detect multiplets (v.0.2.1)[49]. Cells with more than 20% of unique molecular identifiers (UMIs) from mitochondria were filtered out.

To minimize platform and processing differences, raw FASTQ files of public datasets using the Illumina sequencing platform were downloaded and re-processed. Reads were trimmed using Trim Galore (v.0.5.0; http://www.bioinformatics.babraham.ac.uk/projects/trim_galore/) with default parameters before alignment. Trimmed reads were aligned to the same reference as the 10X Genomics single-cell data using STAR[50] (v.2.7.0a). PCR duplicate removal was performed using Picard (v.2.18.25, https://broadinstitute.github.io/picard/). For the dataset from Zhou et al.[31], cell identities (barcodes) were determined based on the first 8 bp on read 2 and UMI deduplication was performed using the 8-bp UMI sequences on read 2 starting at position 9. One mismatch was allowed for cell barcode identification and UMI quantification[51]. Read counts per gene were summarized, using the same gene annotation as the analyses using 10X Genomics datasets, by the featureCounts function from Subread package with a minimal mapping quality requirement of 10[52] (v.1.6.3).

Dimensionality reduction

PHATE visualization. Cells with more than 200 genes detected were included in the analysis. Genes detected in fewer than 30 cells or with fewer than 60 UMIs in total across all single cells were filtered before normalization. Total UMI counts of single cells were normalized to the median UMI counts per cell. Each 10X Genomics RNA-seq library was treated as a different batch. The normalized matrix was square-root-transformed. The default parameters were used for PHATE[27] embedding except the following changes: 'knn=10, decay=15, n_landmark=2,000, t='auto', gamma=0, n_pca=50, mds_solver='smacof' (version 1.0.4).

Data integration. The expression matrices of published datasets and blastoid cells were combined[29]. Cells and genes were filtered with the same standards as described in the PHATE embedding section.

The median-normalized matrix was natural-log-transformed with the addition of a pseudocount of 1. The resulting matrix was zero-centred and scaled per gene (standard deviation is 1) before PCA. The first 50 principal components were calculated using the PCA function from scikit-learn[53] (v.0.23.2).

Batch correction with Harmony. The Harmony[54] algorithm was used to integrate human pre-implantation datasets (Smart-Seq2) with blastoid datasets (10X Genomics), using default parameters (Python implementation v.0.0.5; https://github.com/slowkow/harmonypy). Only batch information was used for correction: all of the Smart-Seq2 libraries from human pre-implantation embryos are one batch, and each 10X Genomics library is one batch.

Uniform manifold approximation and projection. UMAP embedding was performed using batch-corrected principal components as input (v.0.4.6) with the following non-default parameters: 'n_neighbors=20, min_dist=0.5'[55].

Fast interpolation-based t-SNE. t-SNE of single cells was computed using fast interpolation-based t-SNE (FIt-SNE; v.1.2.1)[56]. UMAP was used to reduce the dimensions as described in the UMAP section except n_neighbours was set to 10 and min_dist to 0.1. The resulting matrix was used as initialization for FIt-SNE, with batch-corrected principal components used as input. Perplexity was set to 30, theta to 0.5 and max_iter to 2,000.

Clustering

Single cells were clustered using the louvain algorithm implemented in the Scanpy package[57] (v.1.6.0). In brief, the output of Harmony was used to compute a neighbourhood graph of observations (n_neighbours = 30, method = 'umap', metric = 'euclidean'). Then, the louvain algorithm was used to cluster single cells (resolution = None, flavor = 'vtraag'). For each cluster, if there are more than 10% or 100 cells corresponding to human embryos, we define the blastoid-derived cells as blastocyst-like. Clusters that failed to meet both criteria are unclassified. Heat map visualization of selected genes was performed using ComplexHeatmap (v.2.6.2).

PCA analysis of public datasets

Cell and sample annotations were retrieved from their original publications (see https://github.com/jlduan/Human_blastoid for details). Cells and samples from the same lineage and developmental stage were merged. Lineage specific genes used are from Zhou et al.[31]. Matrices were median-normalized and natural-log-transformed before calling the prcomp function from R's built-in stats package (v.4.0.3). A linear model was used to fit the first two principal components for the human embryo samples using the rlm function from the MASS package (v.7.3.53) in R.

Gene ontology enrichment analysis

Gene ontology (GO) term enrichment was performed using topGO (v.2.40.0; org.Hs.eg.db v.3.11.4). Fisher's exact test was used for enrichment tests (algorithm = 'classic', statistic = 'fisher').

RNA velocity

Spliced and unspliced matrices of reads were summarized using velocyto (v.0.17.17) with default parameters[58]. Low complexity and repeat regions were downloaded from the UCSC browser (on 20 April 2020). scVelo was performed for RNA velocity analysis (v.0.2.2)[59]. The embedding used was computed using FIt-SNE as described in the previous section. The non-default scVelo parameters were: n_neighbours = 30, n_components = 30, velocity_mode = 'dynamical', n_top_genes = 2,000.

Pseudotime construction

Monocle (v.2.18.0) was used for pseudotime analysis, with the UMI matrix used as input. The minimum expression level that constitutes true expression was set to 0.5 and the distribution was set to 'negbinomial.size()'. Genes expressed in more than 30 cells and with more than 60 UMIs captured in all cells were included for pseudotime ordering[60].

Statistical analysis

All quantitative data are presented as mean ± s.d. Experiments were repeated at least three times (repeat number is indicated as *n* in figure legends). Differences between groups were evaluated by Student's *t*-test (two-sided). *P* values are shown in the figures. Graphic analyses were done using GraphPad Prism v.7.0 and 8.0 and Microsoft Excel.

Reporting summary

Further information on research design is available in the Nature Research Reporting Summary linked to this paper.

Data availability

A total of nine single-cell RNA-seq datasets generated in this study have been deposited in the Gene Expression Omnibus (GEO) with accession code GSE150578. Source data are provided with this paper.

Code availability

Code used in this project is provided at https://github.com/jlduan/Human_blastoid.

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Author contributions J.W. and L.Y. conceptualized the idea, designed, analysed and interpreted the experimental results. L.Y. and Y.W. performed most of the human blastoid experiments. M.S. helped with immunofluorescence analysis. D.A.S. designed and prepared DNA constructs. J.D. and G.H. performed scRNA-seq analyses. L.W. performed scRNA-seq experiments. Y.W. and M.S. performed mouse embryo microinjection experiments. K.W. and S.Z. performed all human blastocyst experiments including human blastocyst thaw and culture, immunostaining and cell counting, and performed the grading of human blastocysts and blastoids. J.W. and G.H. supervised the study. L.Y., J.D., G.H. and J.W. wrote the manuscript with input from all authors.

Competing interests The authors declare no competing interests.

Additional information

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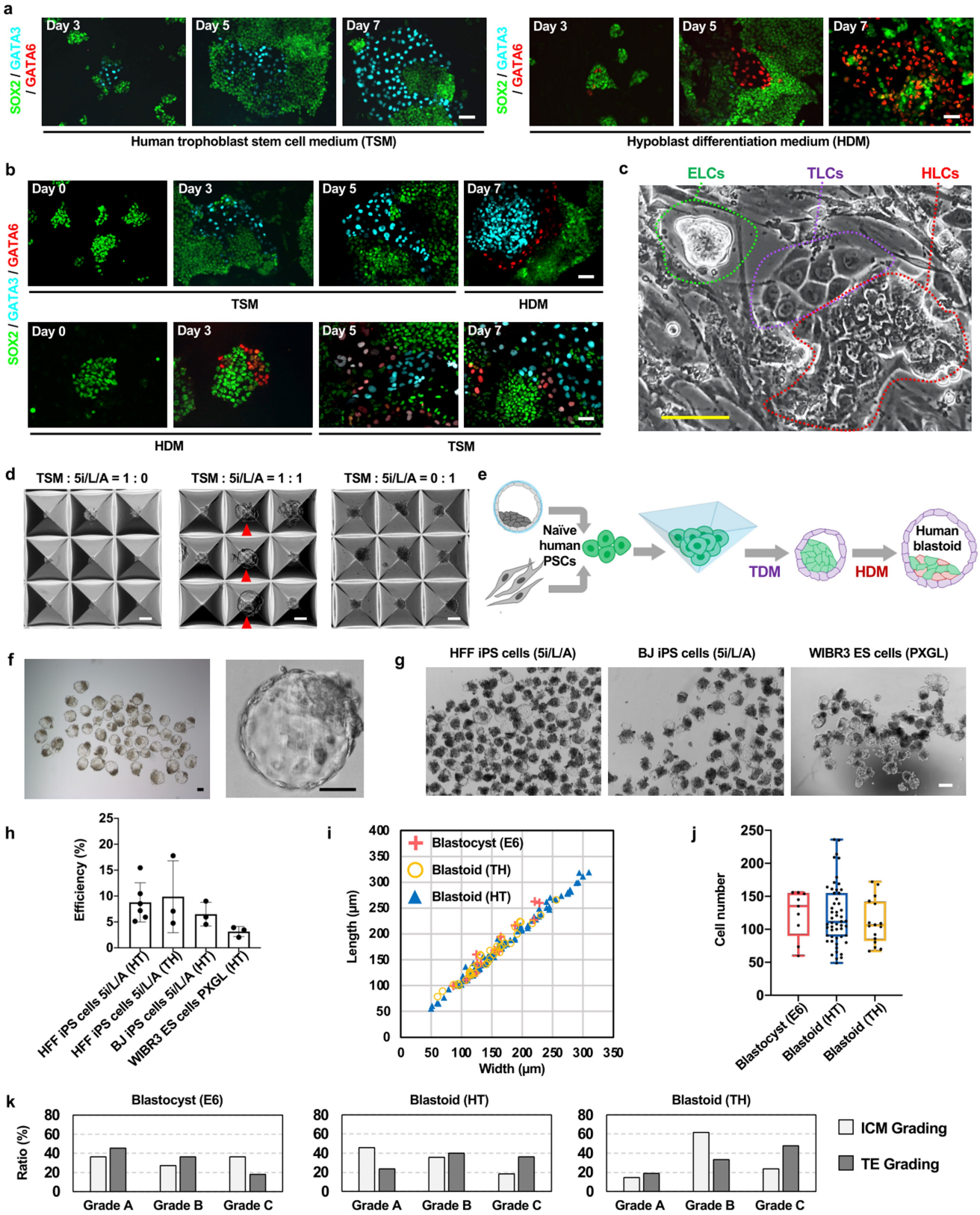
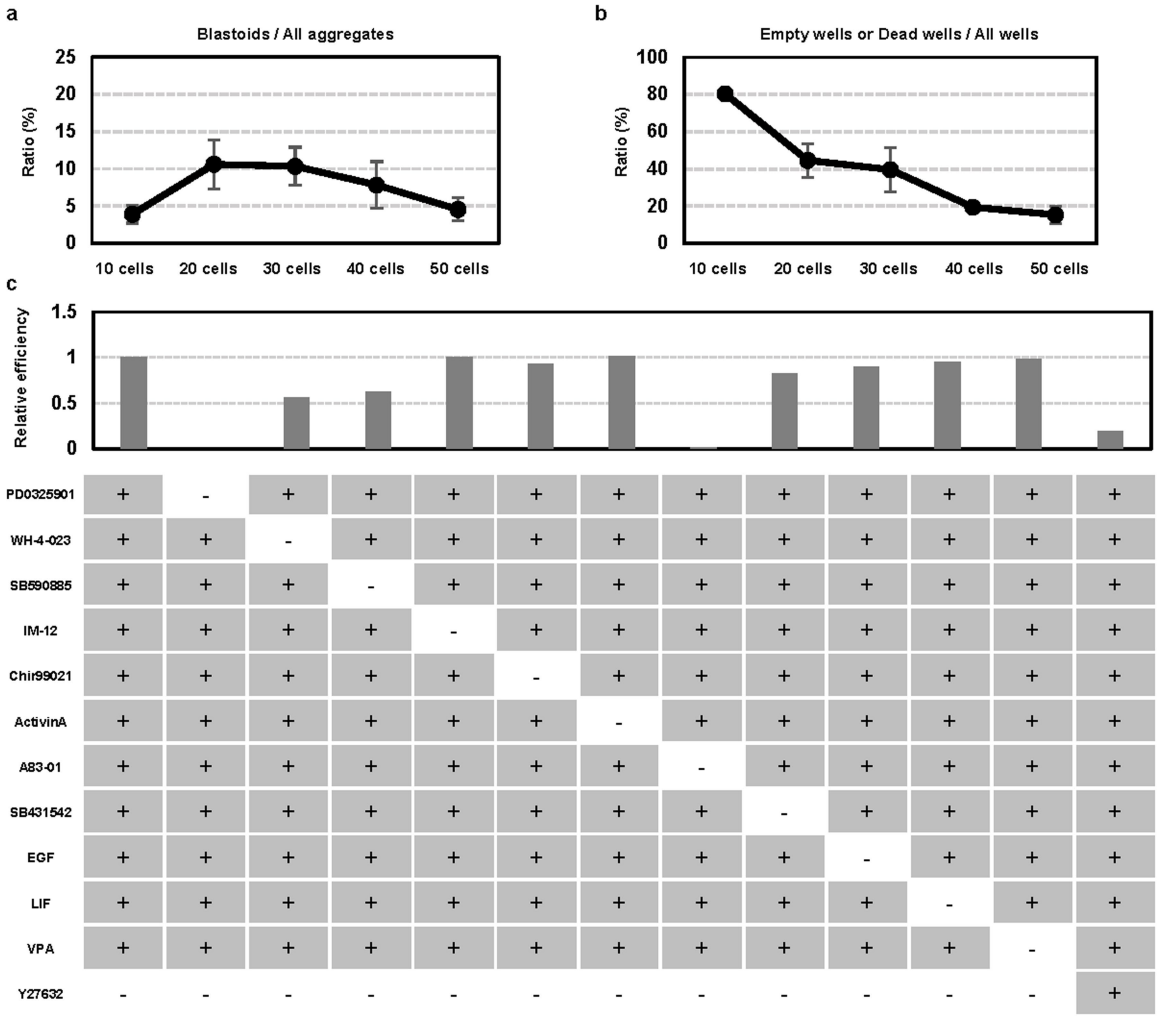
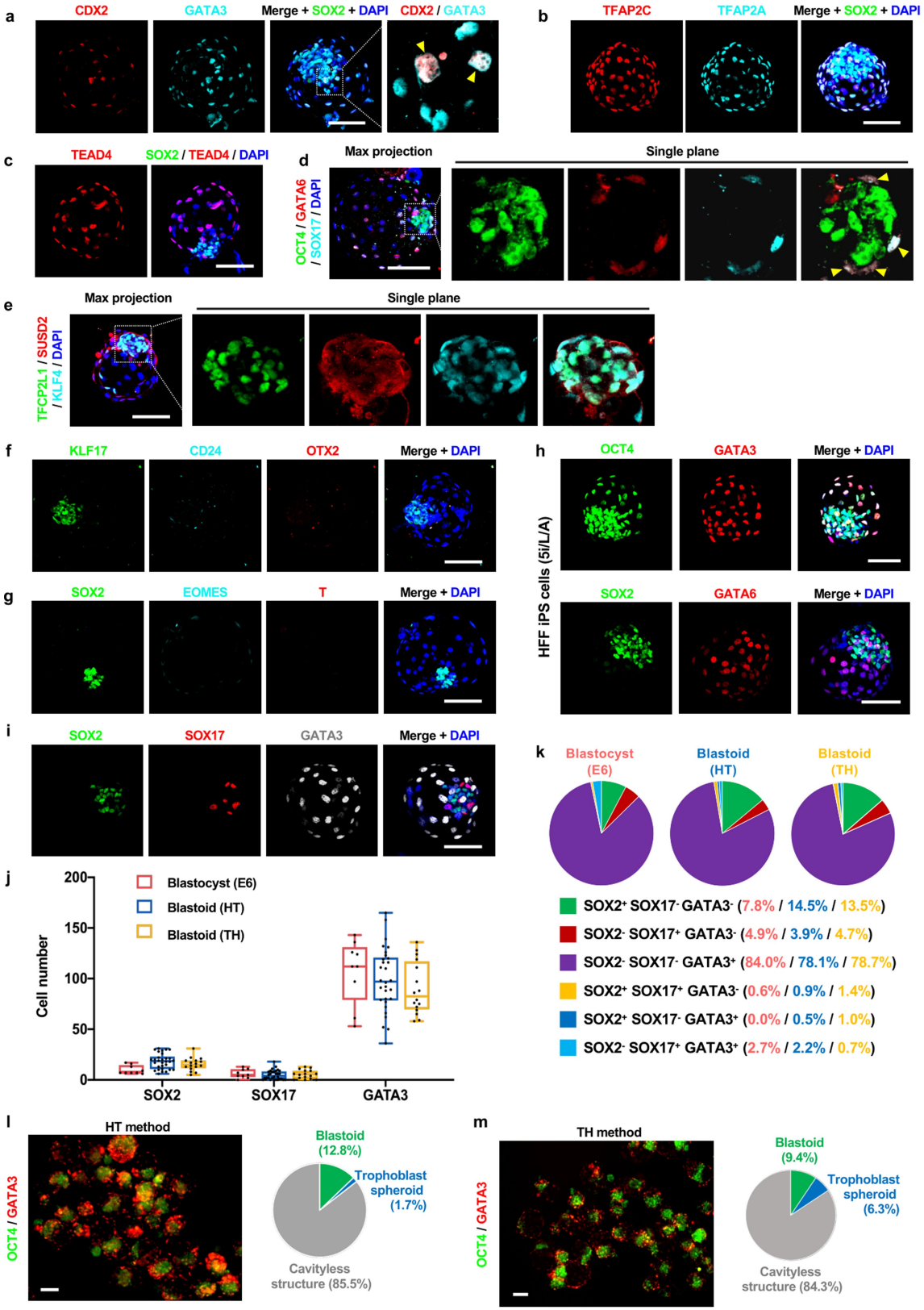


Extended Data Fig. 1 | See next page for caption.

Extended Data Fig. 1 | Generation of human blastoids from naive human PSCs. a, Representative immunofluorescence co-staining images of SOX2, GATA3 and GATA6 showing differentiation of naive human PSCs in human TSM (left) and HDM (right). Scale bars, 100 µm. **b**, Representative immunofluorescence co-staining images of SOX2, GATA3 and GATA6 showing differentiation of naive human PSCs in TSM followed by HDM (top) and HDM followed by TSM (bottom). Scale bars, 100 µm. **c**, A representative phase-contrast image showing the presence of ELCs, HLCs and TLCs in a day-7 codifferentiation culture. Green, purple and red dashed lines encircle a typical colony of ELCs, TLCs and HLCs, respectively. Scale bar, 100 µm. **d**, Representative phase-contrast images showing cell aggregates formed in media with different TSM and 5i/L/A ratios. Red arrowheads indicate blastocyst-like structures. Scale bars, 100 µm. **e**, Schematic of human blastoid formation from naive human PSCs using the TH method. **f**, Representative phase-contrast images of human blastoids (TH method). Scale bars, 100 µm. **g**, Representative phase-contrast images showing human blastoids generated from 5i/L/A-cultured HFF and BJ human iPS cells and PXGL-cultured WIBR3 human ES cells (HT method). Scale bar, 100 µm. **h**, Human blastoid formation efficiencies from 5i/L/A-cultured HFF human iPS cells (HT method, $n = 6$ biological replicates; TH method, $n = 3$ biological replicates), 5i/L/A-cultured BJ human iPS cells (HT method, $n = 3$ biological replicates) and PXGL-cultured WIBR3 human ES cells (HT method, $n = 3$ biological replicates). Data are mean ± s.d. **i**, Lengths and widths of human blastocysts ($n = 13$ biological replicates) and human blastoids (TH method, $n = 36$ biological replicates; HT method, $n = 110$ biological replicates). **j**, Total number of cells per human blastocyst ($n = 9$ biological replicates) and human blastoid (HT method, $n = 52$ biological replicates; TH method, $n = 16$ biological replicates). Box plots show median (centre line), 25th and 75th percentiles (bottom and top of box, respectively), and minimum and maximum values (bottom and top whisker, respectively). **k**, Morphological grading of human blastocysts ($n = 11$ biological replicates) and human blastoids (HT method, $n = 97$ biological replicates; TH method, $n = 21$ biological replicates). TE, trophectoderm. Images in **a**–**d**, **f**, **g** are representative of at least three independent experiments.

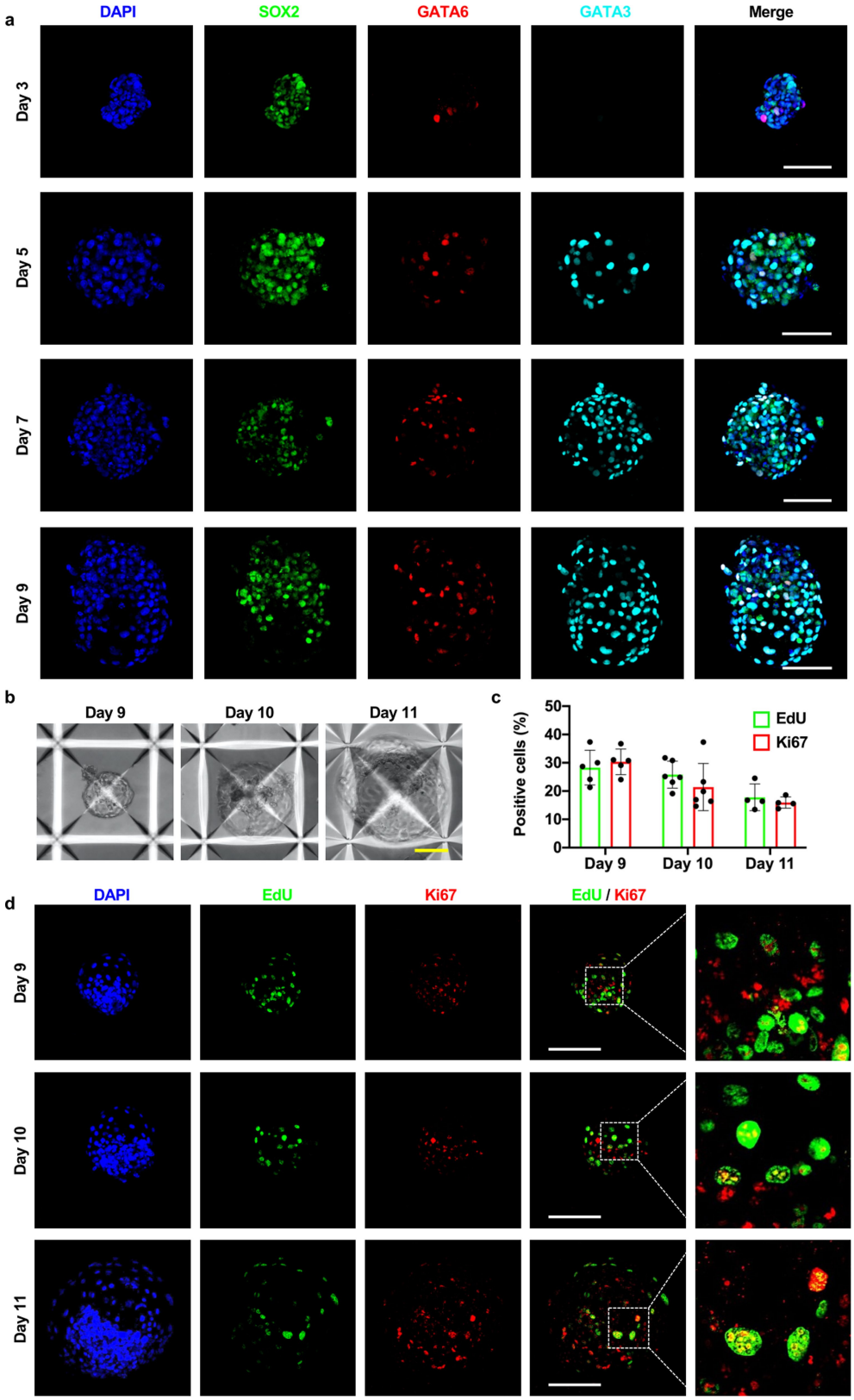


Extended Data Fig. 2 | Optimization of blastoid formation from naive human PSCs. a, Ratio of blastoids to all aggregates generated from different starting cell numbers (HT method, $n = 3$ biological replicates; mean ± s.d.). **b**, Ratio of empty wells or dead wells (wells containing only dead cells) to all wells using different starting cell numbers (HT method, $n = 3$ biological replicates; mean ± s.d.). **c**, The effect of each TDM component on blastoid formation efficiency (HT method).

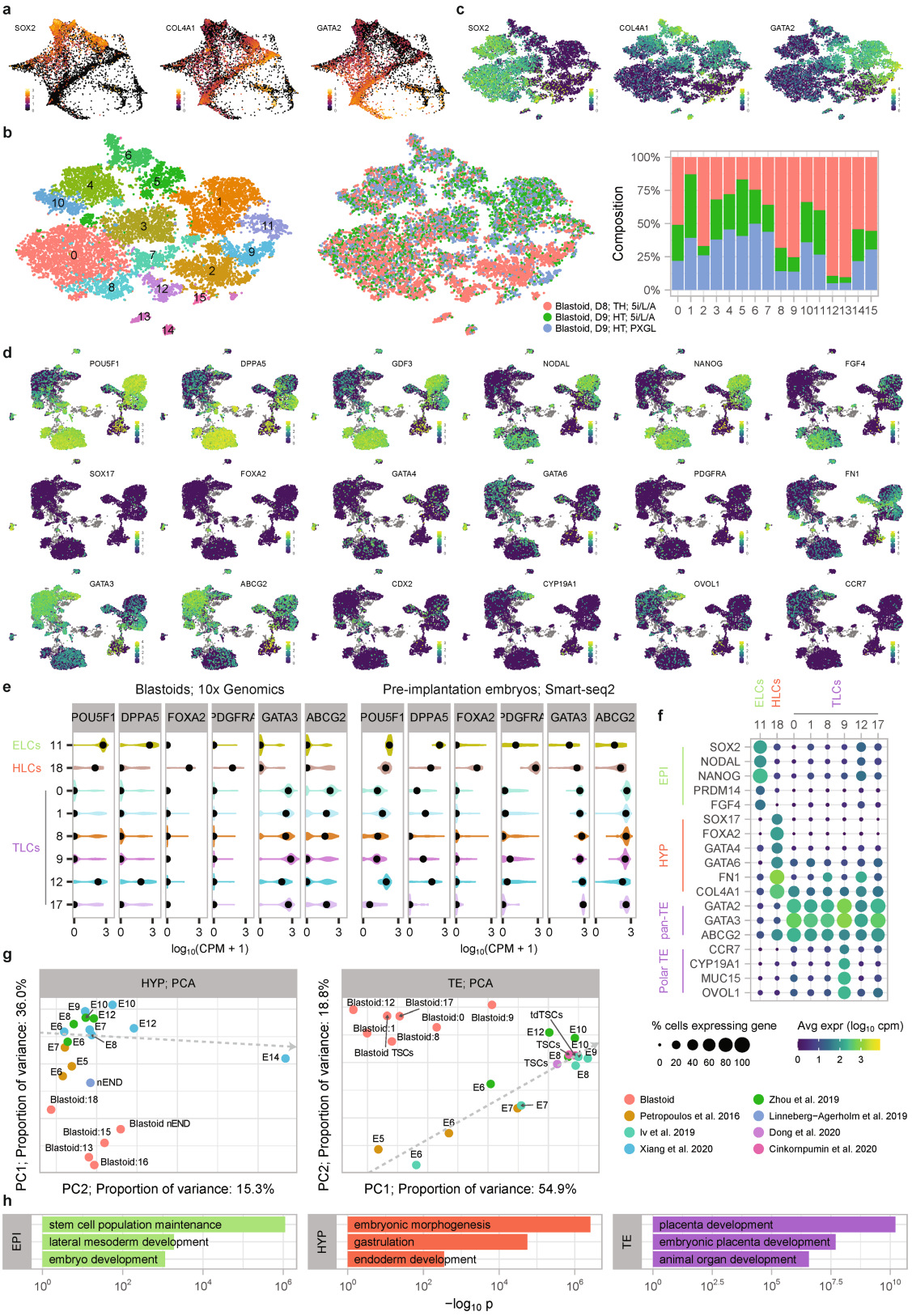


Extended Data Fig. 3 | See next page for caption.

Extended Data Fig. 3 | Cell-lineage composition of human blastoids. **a**, Representative immunofluorescence co-staining images of CDX2, GATA3 and SOX2 in a human blastoid. A higher-magnification image of the boxed area is shown on the right. Yellow arrowheads indicate $CDX2^+GATA3^+$ cells. Scale bar, 100 μm. **b**, Representative immunofluorescence co-staining images of TFAP2A, TFAP2C and SOX2 in a human blastoid. Scale bar, 100 μm. **c**, Representative immunofluorescence co-staining images of TEAD4 and SOX2 in a human blastoid. Scale bar, 100 μm. **d**, Representative immunofluorescence co-staining images of OCT4, GATA6 and SOX17 in a human blastoid. Left, maximum intensity projection of *z*-stack images; right, higher-magnification single-plane images of the boxed area. Yellow arrowheads indicate $GATA6^+SOX17^+$ cells. Scale bar, 100 μm. **e**, Representative immunofluorescence co-staining images of TFCP2L1, SUSD2 and KLF4 in a human blastoid. Left, maximum intensity projection of *z*-stack images; right, higher-magnification single-plane images of the boxed area. Scale bar, 100 μm. **f**, Representative immunofluorescence co-staining images of KLF17, CD24 and OTX2 in a human blastoid. Scale bar, 100 μm. **g**, Representative immunofluorescence co-staining images of SOX2, EOMES and T in a human blastoid. Scale bar, 100 μm. **h**, Representative immunofluorescence co-staining images of GATA3 and OCT4 (top) and GATA6 and SOX2 (bottom) in human blastoids generated from 5i/L/A-cultured HFF iPS cells. Scale bars, 100 μm. **i**, Representative immunofluorescence co-staining images of SOX2, SOX17 and GATA3 in a human blastoid (TH method). Scale bar, 100 μm. **j**, Quantification of the number of $SOX2^+$, $SOX17^+$ and $GATA3^+$ cells in human blastocysts ($n = 9$ biological replicates) and human blastoids (TH method, $n = 16$ biological replicates; HT method, $n = 30$ biological replicates). Box plots show median (centre line), 25th and 75th percentiles (bottom and top of box, respectively), and minimum and maximum values (bottom and top whisker, respectively). **k**, Pie charts showing the frequencies of different expression patterns of SOX2, SOX17 and GATA3 in human blastocysts (left) and human blastoids (TH method, middle; HT method, right). **l**, **m**, Left, representative immunofluorescence co-staining images of OCT4 and GATA3 in cell aggregates generated by the HT (**l**) and TH (**m**) method. Scale bars, 100 μm. Right, percentages of cavityless structures, trophoblast spheroids and blastoids among all aggregates generated by the HT (**l**) and TH (**m**) method. Images in **a**–**i**, **l**, **m** are representative of at least three independent experiments.

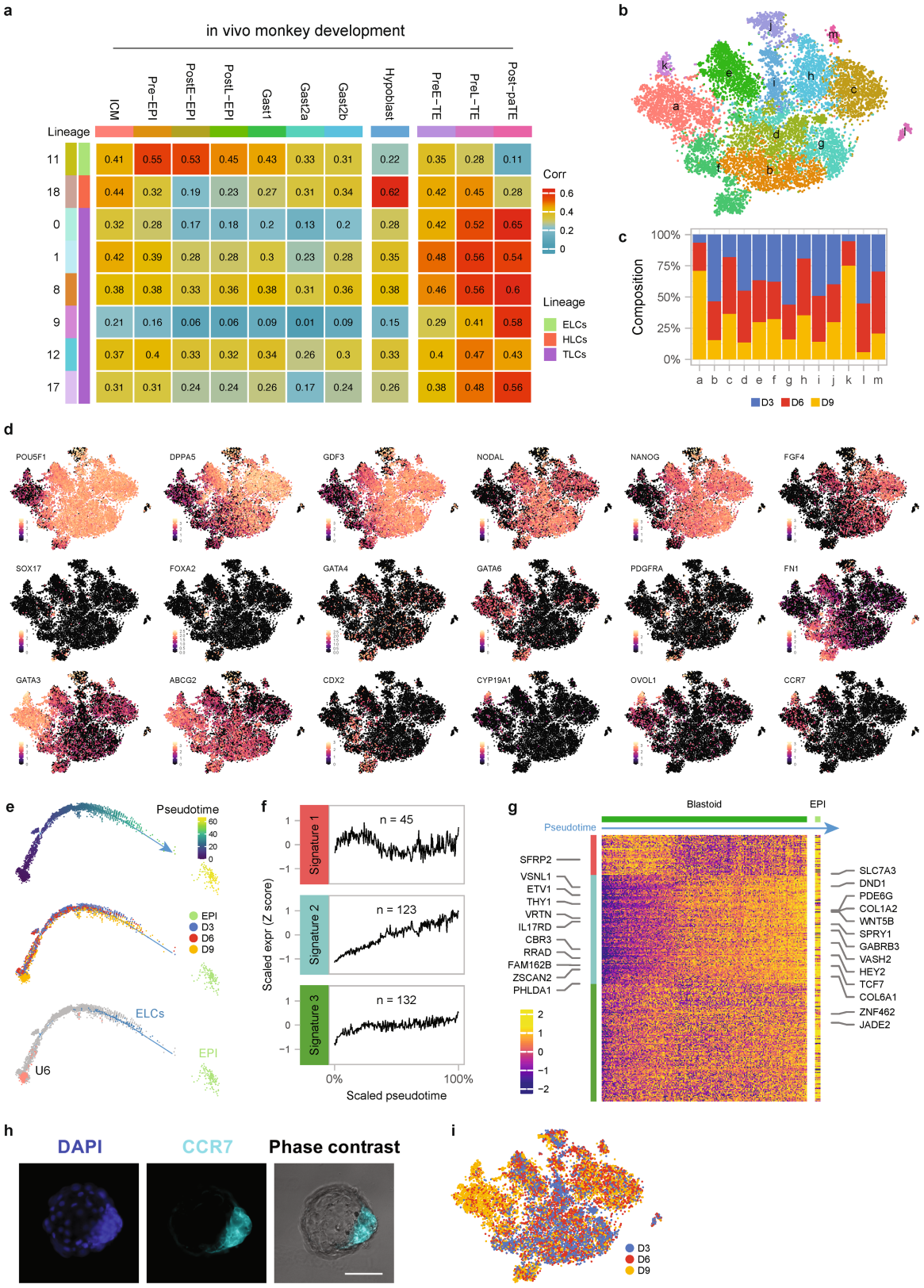


Extended Data Fig. 4 | Time-course studies of lineage differentiation dynamics during human blastoid formation and extended culture of human blastoids. a, Representative immunofluorescence co-staining images of SOX2, GATA6 and GATA3 in cell aggregates generated at different time points during blastoid formation. Scale bars, 100 μm. **b**, Representative phase-contrast images showing extended culture of human blastoids in TDM. Scale bar, 100 μm. **c**, Percentages of EdU^+ or $Ki67^+$ cells in human blastoids after extended culture in TDM (day 9, $n = 5$; day 10, $n = 6$; day 11, $n = 4$ (biological replicates); mean ± s.d.). **d**, Representative immunofluorescence co-staining images of EdU and Ki67 in human blastoids after extended culture in TDM. Higher-magnification images of the boxed areas are shown on the right. Scale bars, 100 μm. Images in **a**, **b**, **d** are representative of at least three independent experiments.



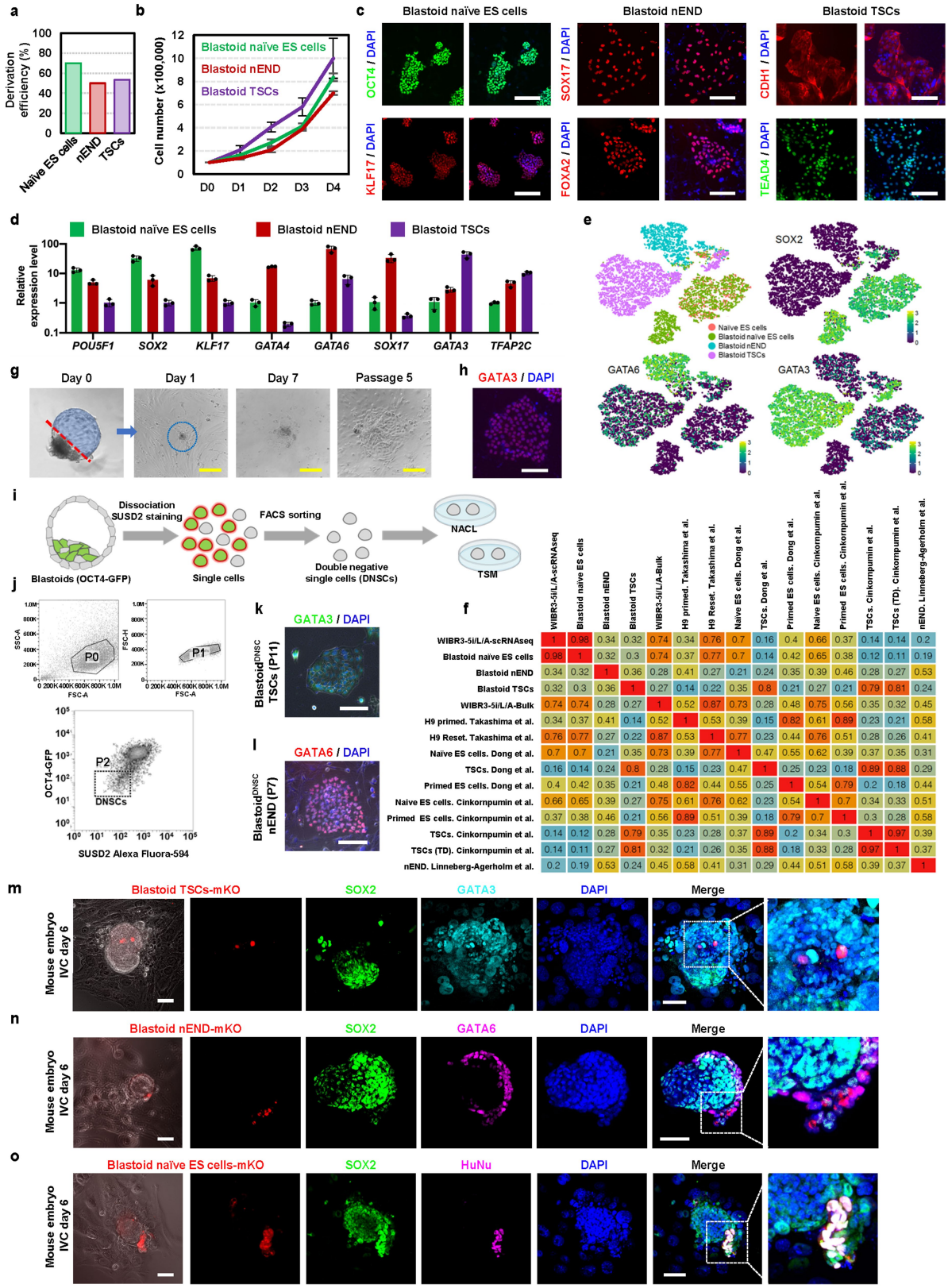
Extended Data Fig. 5 | See next page for caption.

Extended Data Fig. 5 | Single-cell RNA-seq analyses of human blastoids. **a**, PHATE embedding of blastoid cells. Shown is the expression of three lineage markers: *SOX2* (epiblast), *COL4A1* (hypoblast) and *GATA2* (trophectoderm) **b**, Left, t-SNE plots comparing the transcriptomes of blastoid cells from TH and HT derivation protocols. Cluster labels are indicated. Middle, cells are coloured by derivation protocol. 5i/L/A and PXGL indicate two culture conditions for naive human PSCs before blastoid derivation. Right, quantification of the cellular composition of each cluster. **c**, t-SNE plots comparing the transcriptomes of blastoid cells from TH and HT derivation protocols. Shown is the expression of three lineage markers: *SOX2* (epiblast), *COL4A1* (hypoblast) and *GATA2* (trophectoderm). **d**, Joint UMAP embedding of single-cell transcriptomes from blastoids and human pre-implantation embryos (Fig. 2a). Shown is the expression of lineage markers. To emphasize blastoid cells, cells derived from human pre-implantation embryos are coloured grey. **e**, Violin plots illustrating the expression of selected lineage markers from single-cell expression data. Quantified cells are split between blastoids (left) and pre-implantation human embryos (right). **f**, Dot plot indicating the expression of well-known markers of epiblast (EPI), hypoblast (HYP) and trophectoderm (TE). Clusters contain only blastoid cells, and do not contain cells from human embryos. **g**, PCA of blastoid-derived HLCs (left) and TLCs (right) relative to published maps from human embryos and established stem cell lines[19,20,29,31–34]. Embryonic time points and cell line names are indicated (labelled as in each previous study), and the dotted grey lines indicate developmental time axes estimated from human embryo datasets. **h**, Enrichment of GO biological process terms for cluster-specific genes. In **e**–**h**, clusters contain only blastoid cells, and do not contain cells from human embryos.



Extended Data Fig. 6 | See next page for caption.

Extended Data Fig. 6 | Additional single-cell RNA-seq analyses of human blastoids. a, Correlation analysis of blastoid clusters with cells from in vivo cynomolgus monkey embryos (gast, gastrulating cells; postE-EPI, post-implantation early epiblast; postL-EPI, post-implantation late epiblast; post-paTE, post-implantation parietal trophectoderm; pre-EPI, pre-implantation epiblast; preE-TE, pre-implantation early trophectoderm; preL-TE, pre-implantation late trophoectoderm). **b**, Single-cell RNA-seq was performed at days 3, 6 and 9 during blastoid derivation ($n = 3$). t-SNE plot of sequenced cells, coloured by cluster assignment. **c**, Quantification of the cellular composition of clusters in **b**. **d**, Expression of lineage markers from the t-SNE analysis in **b**. **e**, Pseudotime trajectories for the ELCs across a time course of blastoid derivation. Epiblast cells from human pre-implantation embryos are included as a reference. Plots show pseudotime (top), sample time point (middle) and cell identity (bottom). The variability of U6 cells across time points may be due to cellular heterogeneity and unsynchronized differentiation of naive human PSCs during blastoid formation. **f**, **g**, Pseudotime-dependent gene expression signatures for the ELCs. This analysis only includes epiblast-specific genes defined in Zhou et al.[31]. **h**, Representative immunofluorescence and phase-contrast images showing the expression of CCR7 in a human blastoid. Images are representative of at least three independent experiments. Scale bar, 100 µm. **i**, t-SNE plot from Fig. 2h, without velocity analysis.



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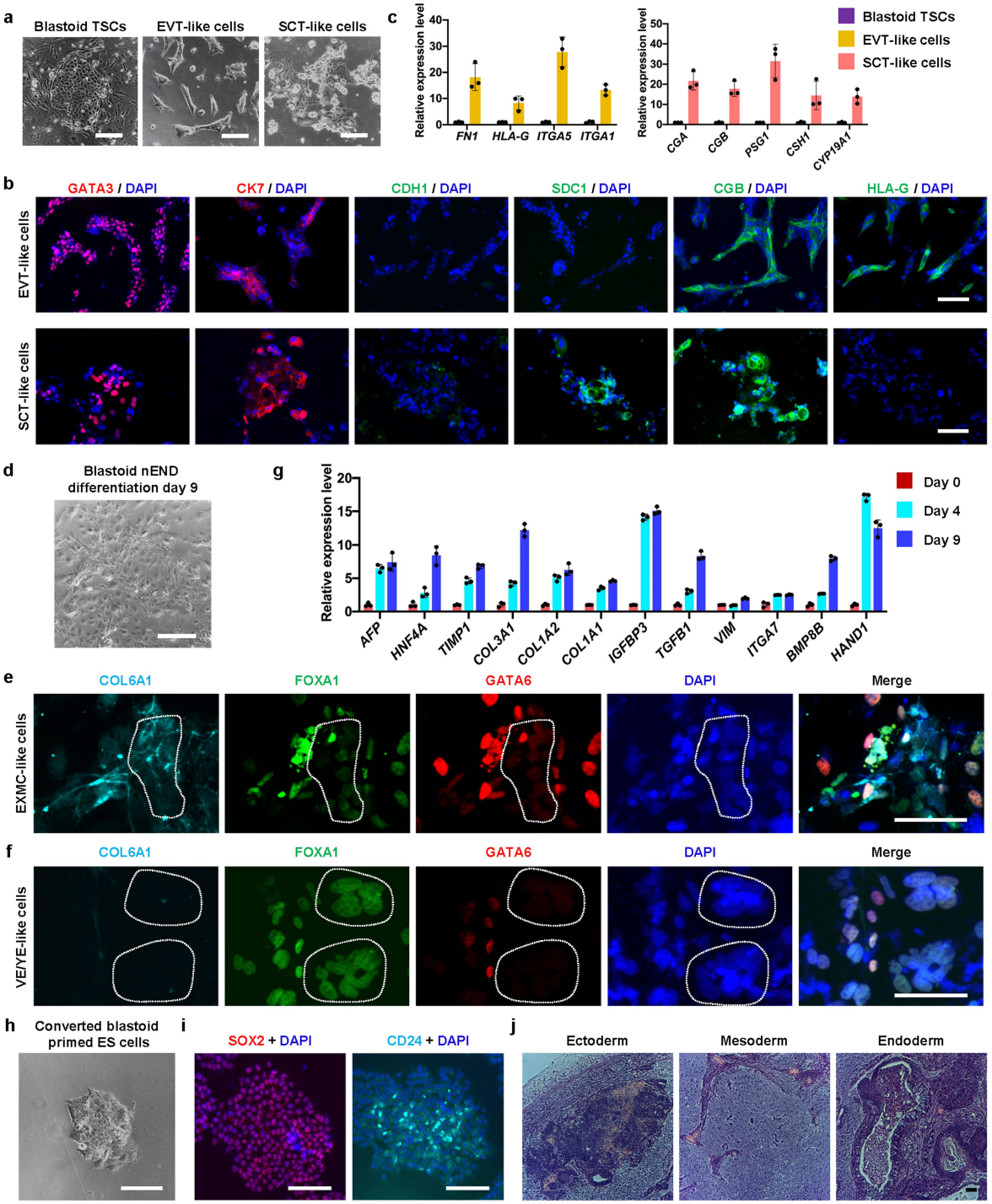
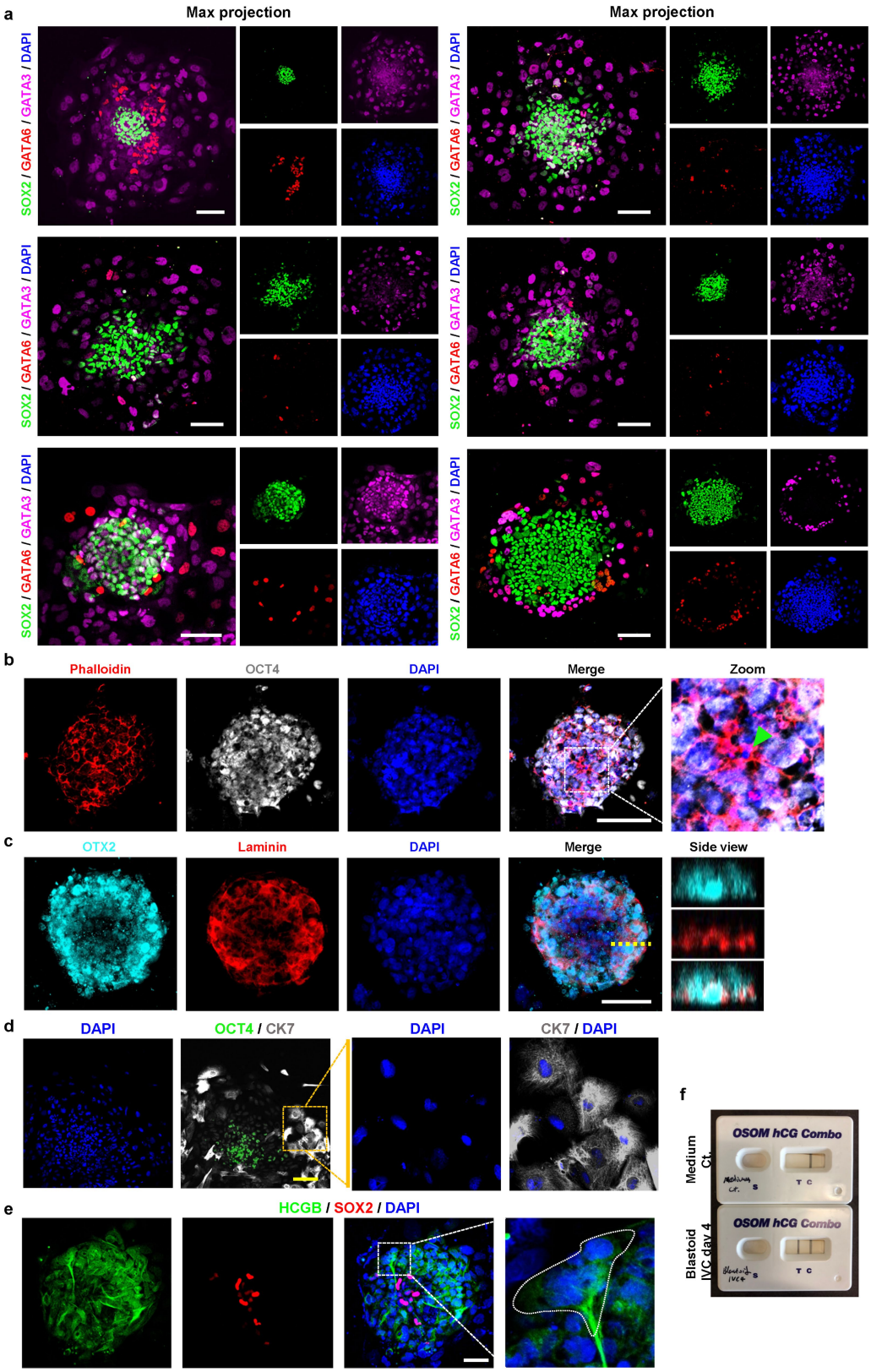
Extended Data Fig. 7 | Derivation of stem cells from human blastoids. **a**, Derivation efficiency for blastoid naive ES cells, blastoid nEND, and blastoid TSCs. Thirty blastoids were used for each. **b**, Growth curves of blastoid naive ES cells, blastoid nEND and blastoid TSCs ($n = 3$ biological replicates; mean ± s.d.). **c**, Representative immunofluorescence images of blastoid naive ES cells (OCT4, KLF17), blastoid nEND (SOX17, FOXA2) and blastoid TSCs (CDH1, TEAD4). Scale bars, 100 µm. **d**, RT–qPCR results showing the relative expression levels of epiblast (*POU5F1*, *SOX2*, *KLF17*), hypoblast (*GATA4*, *GATA6*, *SOX17*) and trophectoderm (*GATA3*, *TFAP2C*) markers in blastoid naive ES cells, blastoid nEND and blastoid TSCs ($n = 3$ biological replicates; mean ± s.d.). **e**, t-SNE of naive ES cells, blastoid naive ES cells, blastoid nEND and blastoid TSCs. Plots are coloured according to sample (top left), SOX2 expression (top right), GATA6 expression (bottom left) and GATA3 expression (bottom right). **f**, Correlation analysis of blastoid naive ES cells, blastoid nEND and blastoid TSCs compared to published transcriptomes[19,20,29,31–34] from established human ES cells, human TSCs and nEND. Each indicated sample represents merged cells (scRNA-seq datasets) or replicates (bulk RNA-seq datasets). The gene sets used in this analysis were defined in Zhou et al.[31]. **g**, Left, representative phase-contrast images showing manual cutting (red dashed line) and outgrowths (days 1 and 7) during the derivation of blastoid TSCs. Right, a representative phase-contrast image showing the morphology of blastoid TSCs (passage 5) derived from manually cut TLCs. Scale bars, 100 µm. **h**, A representative immunofluorescence image showing that human TSCs derived from cut TLCs express GATA3. Scale bar, 100 µm. **i**, Schematic showing the strategy to derive blastoid nEND and blastoid TSCs from OCT4-GFP$^-$ and SUSD2$^-$ double-negative single cells (DNSCs) isolated from human blastoids. **j**, FACS gating and sorting of OCT4-GFP$^-$ and SUSD2$^-$ DNSCs from dissociated human blastoids. **k**, **l**, Representative phase-contrast and immunofluorescence images showing that blastoid TSCs (**k**) and blastoid nEND (**l**) derived from FACS-sorted DNSCs express GATA3 and GATA6, respectively. Scale bars, 100 µm. **m**–**o**, Representative phase-contrast (left) and immunofluorescence (right) images showing that monomeric Kusabira-Orange (mKO)-labelled blastoid TSCs (**m**), blastoid nEND (**n**) and blastoid naive ES cells (**o**) contribute to the extraembryonic ectoderm, visceral endoderm and epiblast compartments of day-6 IVC mouse embryos after blastocyst injections. Higher-magnification images of the boxed areas are shown on the right. Scale bars, 100 µm. Images in **c**, **g**, **h**, **k**–**o** are representative of at least three independent experiments.

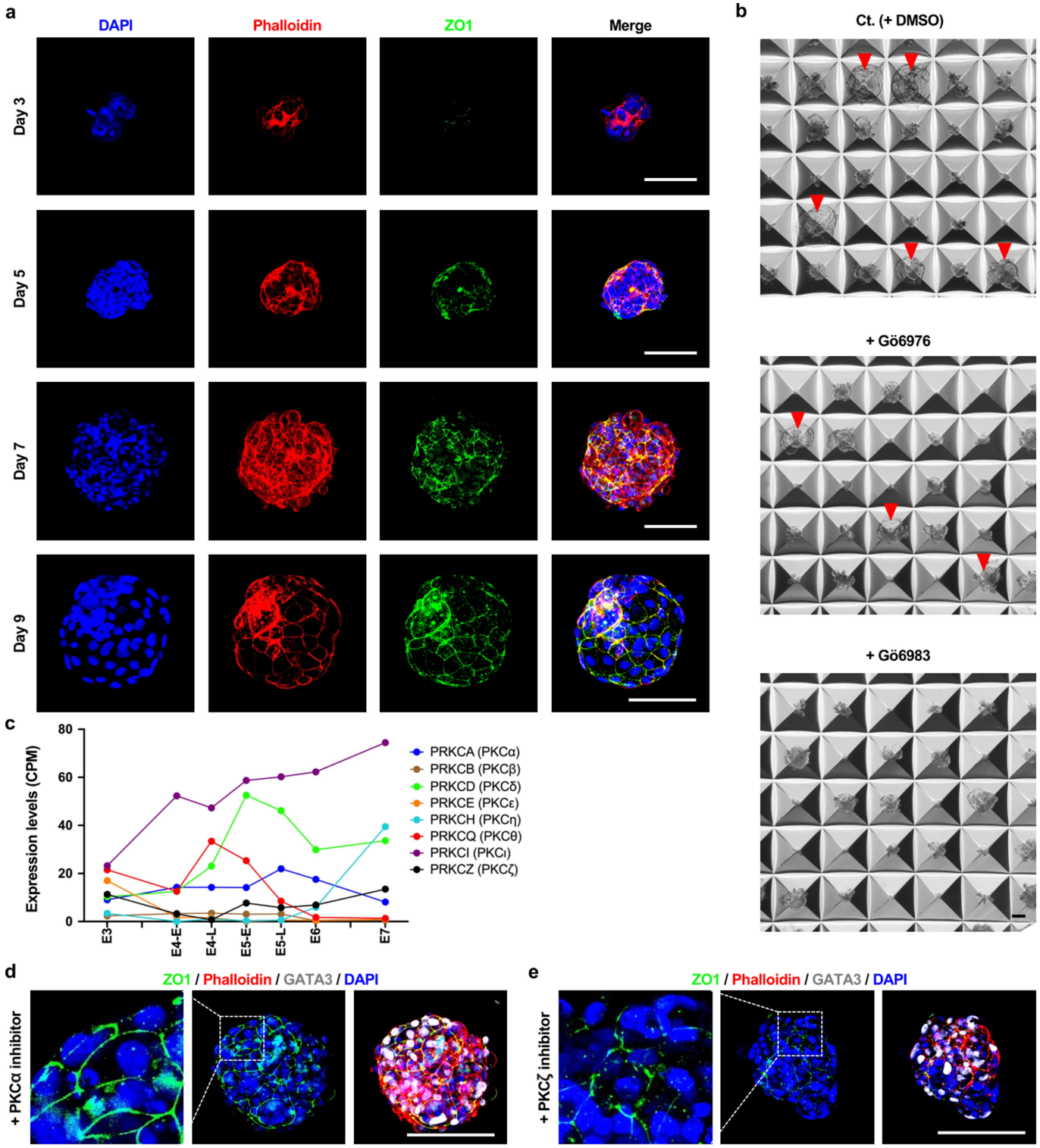


Extended Data Fig. 8 | In vitro differentiation of stem cells derived from human blastoids. a, Representative phase-contrast images showing blastoid TSCs, as well as EVT-like cells and SCT-like cells differentiated from blastoid TSCs. Scale bars, 100 μm. **b**, Representative immunofluorescence images of GATA3, CK7 and CDH1 (trophectoderm markers), SDC1 and CGB (SCT markers) and HLA-G (an EVT marker) in EVT-like cells and SCT-like cells differentiated from blastoid TSCs. Scale bars, 100 μm. **c**, RT–qPCR results showing the relative expression levels of EVT (*FN1*, *HLA-G*, *ITGA5*, *ITGA1*) and SCT (*CGA*, CGB (detects all 6 CGB genes), *PSG1*, *CSH1*, *CYP19A1*) markers in blastoid TSCs, as well as in EVT-like and SCT-like cells differentiated from blastoid TSCs ($n = 3$ biological replicates, mean ± s.d.). **d**, A representative phase-contrast image showing a day-9 differentiation culture of blastoid nEND cells. Scale bar, 100 μm. **e**, **f**, Representative immunofluorescence co-staining images of COL6A1, FOXA1 and GATA6 in a day-9 differentiation culture of blastoid nEND cells. White dashed lines indicate FOXA1$^-$COL6A1$^+$GATA6$^-$ cells (**e**) and FOXA1$^+$COLA6A1$^-$GATA6$^-$ cells (**f**). Scale bars, 100 μm. **g**, RT–qPCR results showing relative expression levels of VE/YE and EXMC markers in day-4 and day-9 differentiation cultures of blastoid nEND cells. ($n = 3$ biological replicates; mean ± s.d.). **h**, A representative phase-contrast image showing the colony morphology of primed human ES cells converted from blastoid naive ES cells. Scale bar, 100 μm. **i**, Representative immunofluorescence images showing that primed human ES cells converted from blastoid naive ES cells stained positive for SOX2 and CD24. Scale bars, 100 μm. **j**, Representative H&E staining images showing that the teratomas formed by primed human ES cells contain tissues from all three embryonic germ layers. Scale bar, 100 μm. Images in **a**, **b**, **d**–**f**, **h**–**j** are representative of at least three independent experiments.



Extended Data Fig. 9 | See next page for caption.

Extended Data Fig. 9 | Culturing human blastoids beyond implantation stages. a, Representative immunofluorescence co-staining images of SOX2, GATA6 and GATA3 in selected day-4 IVC human blastoids, showing abnormal growth of ELCs, TLCs and/or HLCs. Scale bars, 100 µm. **b**, Representative immunofluorescence co-staining images of phalloidin and OCT4 showing a rosette-like structure in a day-4 IVC human blastoid. A higher-magnification image of the boxed area is shown on the right. Green arrowhead indicates the apical domain of the rosette-like structure. Scale bar, 100 µm. **c**, Representative immunofluorescence co-staining images of OTX2 and laminin in a day-4 IVC human blastoid. Side views of the yellow dashed line are shown on the right. Scale bar, 100 µm. **d** Representative immunofluorescence co-staining images of OCT4 and CK7 in a day-4 IVC human blastoid. Higher-magnification images of the boxed area are shown on the right. Scale bar, 100 µm. **e**, Representative immunofluorescence co-staining images of HCGB and SOX2 in a day-4 IVC human blastoid. A higher-magnification image of the boxed area is shown on the right. White dotted line encircles multinucleated cells. Scale bar, 100 µm. **f**, Representative results using a human pregnancy test kit on medium collected from day-4 IVC human blastoids and IVC medium control (Ct.). Images in **a**–**e** are representative of at least three independent experiments.



Extended Data Fig. 10 | PKC isozymes regulate cavity formation in human blastoids. a, Representative immunofluorescence co-staining images of phalloidin and ZO1 in cell aggregates generated at different time points during blastoid formation. Scale bars, 100 µm. **b**, Representative phase-contrast images of cell aggregates after treatments with DMSO control (top), Gö6976 (middle) or Gö6983 (bottom). Red arrowheads indicate blastoids. Scale bar, 100 µm. **c**, Counts per million (CPM) values calculated from a published scRNA-seq dataset[29], showing the expression levels of different PKC isozymes at different stages during human pre-implantation development. E, early stage; L, late stage. **d**, **e**, Representative immunofluorescence co-staining images of ZO1, phalloidin and GATA3 in cell aggregates after treatments with inhibitors of PKCα (**d**) and PKCζ (**e**). Higher-magnification images of the boxed areas are shown on the left. Scale bars, 100 µm. Images in **a**, **b**, **d**, **e** are representative of at least three independent experiments.

nature research

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Reporting Summary

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Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐ A description of all covariates tested
☒	☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☐	☒ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒ Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection

Cultured cell images were taken by Revolve (ECHO). Realtime PCR result were collected by CFX384 system (BIO-RAD). Confocal images were taken by A1R Resonant Scanning Multispectral Confocal Microscope (Nikon) or Leica TCS SP8 confocal microscope (Leica).

RNA-seq data sets were processed and analyzed using the following tools:
Cell Ranger pipeline (version 4.0.0)
Trim Galore (version 0.5.0)
STAR 5 (version 2.7.0a)
Picard (version 2.18.25)
Subread package (version 1.6.3)
PHATE (version 1.0.4)
Scikit-learn (version 0.23.2)
Harmonypy (version 0.0.5)
UMAP (version 0.4.6)
FIt-SNE (version 1.2.1)
SCANPY package (version 1.6.0)
TopGO (version 2.40.0)
org.Hs.eg.db (version: 3.11.4)
velocyto (version 0.17.17)
scVelo (version 0.2.2)
Monocle (version 2.18.0)
R (version 4.0.3)
Python (version 3.8.2)

Code used in this project is provided at https://github.com/jlduan/Human_blastoid.

Data analysis	qPCR data were analyzed by Bio-Rad Maestro 1.0. Statistic analysis and figure were draw by Prism 7 or Microsoft Excel. Immunofluorescence staining images were adjusted by imageJ. RNA-seq data sets were collected and analyzed using the following tools: Cell Ranger pipeline (version 4.0.0) Trim Galore (version 0.5.0) STAR 5 (version 2.7.0a) Picard (version 2.18.25) Subread package (version 1.6.3) PHATE (version 1.0.4) Scikit-learn (version 0.23.2) Harmonypy (version 0.0.5) UMAP (version 0.4.6) FIt-SNE (version 1.2.1) SCANPY package (version 1.6.0) TopGO (version 2.40.0) org.Hs.eg.db (version: 3.11.4) velocyto (version 0.17.17) scVelo (version 0.2.2) Monocle (version 2.18.0) ComplexHeatmap (2.6.2) R (version 4.0.3) Python (version 3.8.2)

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors/reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

A total of 9 single cell RNA-seq datasets generated in this study have been deposited in GEO: GSE150578.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Sample size sufficiency was determined by previous experiments from our laboratory. Preliminary experiments were performed when possible to determine requirements for sample size, taking into account resources available and ethical, reductionist animal use.
Data exclusions	No data were excluded from the analysis
Replication	All experiments were replicated or performed independently for at least three times, and all attempts at replication were successful.
Randomization	To generate blastoids, naive hPSCs were counted and seeded into Aggrewell, and randomly allocated into each microwell. All generated blastoid were randomly picked and used for the following experiments: cell number counting, width and length measurements, grading, immunofluorescence analysis, stem cell lines derivation, chemical inhibitor treatments and prolonged culture in IVC media. All mice used for chimera analysis were randomly allocated to the experimental groups. For experiments other than those mentioned here, random allocation is not relevant. For example, for stem cell differentiation experiments the investigators need to know specific stem cell type, plating density, differentiation protocol and differentiation time and therefore random allocation was not performed for these experiments.
Blinding	To process experimental samples for downstream analysis, the Investigators needed to know the information regarding pluripotency state, culture condition, genetic background, stem cell types and which method/protocol to use and therefore were not blinded. Data collection and analysis (e.g. grading, immunofluoresence analysis, cell number counting, blastoids/blastocyst width and length measurements) were blinded.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology
☐	☒ Animals and other organisms
☐	☒ Human research participants
☒	☐ Clinical data

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

Oct-3/4 (C-10) antibody (Santa Cruz Biotechnology)
Sox-2 (E-4) antibody (Santa Cruz Biotechnology)
Human SOX17 Affinity Purified Polyclonal Ab antibody (R&D Systems)
Recombinant Anti-GATA3 antibody [EPR16651] - ChIP Grade (Abcam)
Human GATA-6 Affinity Purified Polyclonal Ab antibody (R&D Systems)
Sox-17 (3.5CH) antibody (Santa Cruz Biotechnology)
Anti-KLF17 antibody (Atlas antibodies)
Rabbit anti-OCT4 (Abcam)
Keratin 7 (D1E4) XP Rabbit mAb antibody (Cell Signaling Technology)
GATA3-human antibody (Santa Cruz Biotechnology)
Anti-SOX2 antibody (Abcam)
Anti-ZO-1 Monoclonal Antibody (Innovate Research)
Anti-CDX-2, Clone CDX2-88 antibody (BioGenex)
hCG beta antibody [5H4-E2] (Abcam)
Rabbit Anti-Laminin Polyclonal Antibody (Abcam)
Rabbit Anti-Collagen VI Polyclonal Antibody (Abcam)
FOXA1 antibody (Abcam)
Mouse KLF4 Affinity Purified Polyclonal Ab antibody (R&D Systems)
Human TFCP2L1 Affinity Purified Polyclonal Ab antibody (R&D Systems)
Human Otx2 Affinity Purified Polyclonal Ab antibody (R&D Systems)
Mouse Anti-TEAD4 Monoclonal Antibody (Abcam)
CCR7 Recombinant Rabbit Monoclonal Antibody (SR36-04) (Invitrogen)
ZO1 tight junction protein antibody (Abcam)
HLA-G (4H84) antibody (Santa Cruz Biotechnology)
Syndecan-1 (DL-101) antibody (Santa Cruz Biotechnology)
Purified Mouse Anti-Ki-67 Clone B56 antibody (BD biosciences)
TBR2 / Eomes antibody (Abcam)
AP-2alpha (3B5) antibody (Santa Cruz Biotechnology)
AP-2gamma (6E4/4) antibody (Santa Cruz Biotechnology)
Anti-Brachyury / Bry antibody [EPR18113] (Abcam)
FoxA2/HNF3-beta Antibody (Cell Signaling Technology)
Anti-Human Nuclear Antigen antibody (Abcam)
E-Cadherin (24E10) Rabbit mAb antibody (Cell Signaling Technology)
PKC delta (G-9) antibody (PKCδ) (Santa Cruz Biotechnology)
PKC zeta (H-1) antibody (PKCζ) (Santa Cruz Biotechnology)
PKC eta Polyclonal Antibody (PKCη) (Thermo Fisher Scientific)
APC/Cyanine7 anti-human CD24 antibody (BioLegend)
PE anti-human SUSD2 antibody (BioLegend)
Alexa Fluor® 594 Phalloidin antibody (Invitrogen)
Donkey anti rabbit IgG (H+L) 647 (Invitrogen)
Donkey anti rabbit IgG (H+L) 594 (Invitrogen)
Donkey anti mouse IgG (H+L) 488 (Invitrogen)
Donkey anti mouse IgG (H+L) 594 (Invitrogen)
Donkey anti goat IgG (H+L) 647 (Invitrogen)
Donkey anti goat IgG (H+L) 594 (Invitrogen)
Goat anti rabbit IgG (H+L) 594 (Invitrogen)
Goat anti rabbit IgG (H+L) 488 (Invitrogen)
Goat anti-Mouse IgG 1, Alexa Fluor 647 (Invitrogen)
Goat anti-Mouse IgG 1, Alexa Fluor 488 (Invitrogen)
Goat anti-Mouse IgG2b, Alexa Fluor 594 (Invitrogen)
Goat anti-Mouse IgG2b, Alexa Fluor 488 (Invitrogen)
Goat anti-Mouse IgG2a, Alexa Fluor 488 (Invitrogen)
Goat anti-Mouse IgG2a, Alexa Fluor 594 (Invitrogen)

Validation

- anti-OCT-3/4 antibody (sc-5279, Santa Cruz Biotechnology): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 1858 publications. https://www.scbt.com/p/oct-3-4-antibody-c-10?

productCanUrl=oct-3-4-antibody-c-10&_requestid=4187273
- anti-SOX2 (E-4) antibody (sc-365823, Santa Cruz Biotechnology): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 128 publications. https://www.scbt.com/p/sox-2-antibody-e-4?productCanUrl=sox-2-antibody-e-4&_requestid=4187791
- anti-SOX17 antibody (MAB1924, R&D Systems): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 10 publications. https://www.rndsystems.com/products/human-sox17-antibody-245013_mab1924
- anti-GATA3 antibody [EPR16651] - ChIP Grade (ab199428, Abcam): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 17 publications.
https://www.abcam.com/gata3-antibody-epr16651-chip-grade-ab199428.html
- anti GATA-6 antibody (AF1700, R&D Systems): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 32 publications.
https://www.rndsystems.com/cn/products/human-gata-6-antibody_af1700#product-citations
- anti Sox-17 (3.5CH) antibody (sc-130295, Santa Cruz Biotechnology): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 5 publications. https://www.scbt.com/p/sox-17-antibody-3-5ch?productCanUrl=sox-17-antibody-3-5ch&_requestid=988191
- anti-KLF17 antibody (HPA024629, ATLAS): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 7 publications. https://www.atlasantibodies.com/products/antibodies/primary-antibodies/triple-a-polyclonals/klf17-antibody-hpa024629/
- anti-OCT4 antibody (ab137427, Abcam): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 9 publications.
https://www.abcam.com/oct4-antibody-ab137427.html
- anti-Keratin 7 antibody (4465, Cell Signaling Technology): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 11 publications.
https://www.cellsignal.com/products/primary-antibodies/keratin-7-d1e4-xp-rabbit-mab/4465?Ntk=Products&Ntt=4465
- anti-GATA3 antibody (HG3-31) (sc-268, Santa Cruz Biotechnology): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 384 publications.
https://www.scbt.com/p/gata-3-antibody-hg3-31?productCanUrl=gata-3-antibody-hg3-31&_requestid=989141
- Anti-SOX2 antibody (ab137385, Abcam): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 12 publications.
https://www.abcam.com/sox2-antibody-ab137385.html
- Anti-ZO-1 Antibody (33-9100, Innovate Research): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 599 publications.
https://www.thermofisher.com/antibody/product/ZO-1-Antibody-clone-ZO1-1A12-Monoclonal/33-9100
- Anti-CDX-2 antibody (MU392A-UC, BioGenex): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 49 publications. https://www.labome.com/product/Biogenex/MU392A-UC.html
- anti-hCG beta antibody (ab9582, Abcam): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 11 publications.
https://www.abcam.com/hcg-beta-antibody-5h4-e2-ab9582.html
- Anti-Laminin antibody (ab11575, Abcam): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 357 publications.
https://www.abcam.com/laminin-antibody-ab11575.html
- Anti-Collagen VI antibody (ab6588, Abcam): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 91 publications.
https://www.abcam.com/collagen-vi-antibody-ab6588.html
- Anti-FOXA1 antibody (ab55178, Abcam): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 10 publications.
https://www.abcam.com/foxa1-antibody-ab55178.html
- Anti-SOX2 antibody (AF3158, R&D Systems): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 33 publications.
https://www.rndsystems.com/cn/products/mouse-klf4-antibody_af3158
- anti-TFCP2L1 antibody (AF5726, R&D Systems): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 7 publications.
https://www.rndsystems.com/cn/products/human-tfcp2l1-antibody_af5726
- anti-Otx2 antibody (AF1979, R&D Systems): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 37 publications.
https://www.rndsystems.com/cn/products/human-otx2-antibody_af1979
- Anti-TEAD4 antibody (ab58310, Abcam): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 47 publications.
https://www.abcam.com/tead4-antibody-ab58310.html
- anti-CCR7 Antibody (SR36-04) (MA5-31992, Invitrogen): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 0 publications.
https://www.thermofisher.com/antibody/product/CCR7-Antibody-clone-SR36-04-Recombinant-Monoclonal/MA5-31992
- anti-ZO1 antibody (ab965875, Abcam): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 43 publications.
https://www.abcam.com/zo1-tight-junction-protein-antibody-ab96587.html
- anti-HLA-G antibody (sc-21799, Santa Cruz Biotechnology): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 54 publications.
https://www.scbt.com/p/hla-g-antibody-4h84?productCanUrl=hla-g-antibody-4h84&_requestid=992820
- anti-Syndecan-1 antibody (sc-12765, Santa Cruz Biotechnology): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 61 publications.
https://www.scbt.com/p/syndecan-1-antibody-dl-101?productCanUrl=syndecan-1-antibody-dl-101&_requestid=992965
- anti-Ki-67 antibody (550609, BD biosciences): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 14 publications.
https://www.bdbiosciences.com/us/applications/research/intracellular-flow/intracellular-antibodies-and-isotype-controls/anti-

rat-antibodies/purified-mouse-anti-ki-67-b56/p/550609
- anti-TBR2 / Eomes antibody (ab23345, Abcam): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 323 publications.
https://www.abcam.com/tbr2--eomes-antibody-ab23345.html
- anti-AP-2 alpha (3B5) antibody (sc-12726, Santa Cruz Biotechnology): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 68 publications.
https://www.scbt.com/p/ap-2alpha-antibody-3b5?productCanUrl=ap-2alpha-antibody-3b5&_requestid=993601
- anti-AP-2 gamma (6E4/4) antibody (sc-12762, Santa Cruz Biotechnology): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 80 publications.
https://www.scbt.com/p/ap-2gamma-antibody-6e4-4?productCanUrl=ap-2gamma-antibody-6e4-4&_requestid=993769
- anti-Brachyury / Bry antibody [EPR18113] (ab209665, Abcam): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 6 publications.
https://www.abcam.com/brachyury--bry-antibody-epr18113-ab209665.html
- anti-FoxA2/HNF3-beta Antibody (3143, Cell Signaling Technology): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 15 publications. This product has been discontinued.
- anti-Human Nuclear Antigen antibody (ab191181, Abcam): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 23 publications.
https://www.abcam.com/human-nuclear-antigen-antibody-235-1-ab191181.html
- anti-E-Cadherin (24E10) antibody (3195, Cell Signaling Technology): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 1260 publications.
https://www.cellsignal.com/products/primary-antibodies/e-cadherin-24e10-rabbit-mab/3195
- anti-PKC delta (G-9) antibody (PKCδ) (sc-8402, Santa Cruz Biotechnology): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 26 publications.
https://www.scbt.com/p/pkc-delta-antibody-g-9
- anti-PKC zeta (H-1) antibody (PKCζ) (sc-17781, Santa Cruz Biotechnology): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 105 publications.
https://www.scbt.com/p/pkc-zeta-antibody-h-1?productCanUrl=pkc-zeta-antibody-h-1&_requestid=994739
- anti-PKC eta Polyclonal Antibody (PKCη) (PA5-84168, Thermo Fisher Scientific): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 0 publications.
https://www.thermofisher.com/antibody/product/PKC-eta-Antibody-Polyclonal/PA5-84168
- anti-human CD24 antibody (311131, BioLegend): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 4 publications.
https://www.biolegend.com/en-us/search-results/apc-cyanine7-anti-human-cd24-antibody-12387
- anti-human SUSD2 antibody (327406, BioLegend): The antibody guarantee covers the use of the antibody for IF applications. The antibody has been referenced in 7 publications.
https://www.biolegend.com/en-us/products/pe-anti-human-susd2-antibody-4354
- anti-human SUSD2 antibody (327406, BioLegend): The antibody guarantee covers the use of the antibody for IF and flow cytometry applications. The antibody has been referenced in 7 publications.
https://www.biolegend.com/en-us/products/pe-anti-human-susd2-antibody-4354

Eukaryotic cell lines

Policy information about cell lines

Cell line source(s)	Primed HFF-hiPSCs and BJ-hiPSCs were generated using episomal vectors from human foreskin fibroblasts (HFF, ATCC, CRL-2429) and BJ fibroblasts (ATCC CRL-2522), which were reported previously (Wu et al., Nature, 2015; Wu et al., Cell, 2017), and converted to the 5iLA culture condition. 5iLA-cultured WIBR3 hESCs (OCT4-ΔPE-EGFP) were reported previously (Theunissen et al., Cell stem cell, 2014) and kindly provided by Rudolf Jaenisch (MIT) and Thorold Theunissen (Washington University)
Authentication	Cell lines were authenticated by genomic PCR, RT-qPCRs, immunostaining, RNA-seq and/or in vitro differentiation.
Mycoplasma contamination	Mycoplasma contamination was routinely checked and negative results were obtained.
Commonly misidentified lines (See ICLAC register)	No commonly misidentified cell lines were used in the study.

Animals and other organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research

Laboratory animals	Mice strain: CD-1 (ICR) and NOD-SCID. Blastocysts collected from 3–4 months old female CD-1 (ICR) mice were used for microinjection. Both male and female immunodeficiency NOD-SCID mice (~10 weeks old) were used for teratoma assays. Mice were housed in 12-hr light/12-hr dark cycle 22.1–22.3 °C and 33–44% humidity.
Wild animals	No wild animals were used in the study.
Field-collected samples	No field collected samples were used in the study.
Ethics oversight	UT Southwestern Institutional Animal Care and Use Committee (IACUC)

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Human research participants

Policy information about studies involving human research participants

Population characteristics	Couple A (28 years old male and 27 years old female) were clinically diagnosed as secondary sterility. Long protocol was used for controlled ovarian hyperstimulation. Conventional in vitro fertilization (IVF) was chosen as the method of fertilization. The donated embryos used in this study were cryopreserved blastocysts. Couple B (33 years old male and 32 years old female) were clinically diagnosed as secondary sterility. Long protocol was used for controlled ovarian hyperstimulation. Conventional IVF was chosen as the method of fertilization. The donated embryos used in this study were cryopreserved blastocysts. Couple C (34 years old male and 25 years old female) were clinically diagnosed as secondary sterility. Long protocol was used for controlled ovarian hyperstimulation. Conventional IVF was chosen as the method of fertilization. The donated embryos used in this study were cryopreserved blastocysts.
Recruitment	Selection criteria: (1) Both husband and wife are less than 35 years old; (2) Chromosome test normal; (3) Successfully delivered at least one healthy baby after IVF/ICSI-ET treatment. (4) voluntarily gave up to keep the remaining frozen embryos. and (5) voluntarily donated embryos for medical research. There are no self-selection bias or other biases that are likely to impact results.
Ethics oversight	All human embryo related work was conducted at Kunming Medical University. The studies involving human embryos were approved by the Medical Ethical Committee of Kunming Medical University. Medical Ethical Committee of Kunming Medical University was established by Kunming Medical University and conducts Independent, competent and timely review and supervision of scientific research projects involving human research participants in Kunming Medical University and the affiliated hospitals. All human ESC/iPSC and human blastoid work was conducted at UT Southwestern Medical Center, and reviewed and approved by the UT Southwestern Stem Cell Oversight Committee (SCRO) [Registration #29]. All human-mouse ex vivo interspecies chimeric experimental studies were reviewed and approved by the UT Southwestern Stem Cell Oversight Committee (SCRO) [Registration #14].

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Flow Cytometry

Plots

Confirm that:

☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).

☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).

☒ All plots are contour plots with outliers or pseudocolor plots.

☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	Cells were dissociated into single cells by TrypLE and fixed by 4% PFA
Instrument	FACScalibur system, BD
Software	Analyzed by Flowjo
Cell population abundance	Sample purity was verified as necessary
Gating strategy	Gating strategies are shown in the figures (Extended Data Fig. 7j). Preliminary FSC/SCC gates were used to remove debris, doublets and other aggregated particles from the cell population. Boundaries between positive and negative staining cell populations are defined by using unstained or unlabeled negative control. The SUSD2 and GFP double negative cells were sorted.

☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.